

Phenix Overview and General Reference (Part 1 of 5)

20 documentation pages

Generated from local Phenix documentation

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Phenix Documentation

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Phenix Documentation - version dev-svn

NEW:Phenix documentation for the AI Agent for AI-assisted structure determination, the Phenix chatbot, and AI analysis of results Crystallography Cryo-EM Predicted models How to install GUI introduction FAQs Video tutorials Tutorials & Examples Overview of programs Other links Bibliography Dictionary Index Find a program or regression test Citing Phenix If you use PHENIX please cite: Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. D. Liebschner, P. V. Afonine, M. L. Baker, G. Bunkóczy, V. B. Chen, T. I. Croll, B. Hintze, L.-W. Hung, S. Jain, A. J. McCoy, N. W. Moriarty, R. D. Oeffner, B. K. Poon, M. G. Prisant, R. J. Read, J. S. Richardson, D. C. Richardson, M. D. Sammito, O. V. Sobolev, D. H. Stockwell, T. C. Terwilliger, A. G. Urzhumtsev, L. L. Videau, C. J. Williams and P. D. Adams. *Acta Cryst.* (2019). D75, 861-877

Other links

Citing Phenix

If you use PHENIX please cite:

Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. D. Liebschner, P. V. Afonine, M. L. Baker, G. Bunkóczy, V. B. Chen, T. I. Croll, B. Hintze, L.-W. Hung, S. Jain, A. J. McCoy, N. W. Moriarty, R. D. Oeffner, B. K. Poon, M. G. Prisant, R. J. Read, J. S. Richardson, D. C. Richardson, M. D. Sammito, O. V. Sobolev, D. H. Stockwell, T. C. Terwilliger, A. G. Urzhumtsev, L. L. Videau, C. J. Williams and P. D. Adams. *Acta Cryst.* (2019). D75, 861-877

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What is PHENIX

[what-is-phenix.html](#)

The PHENIX software suite is a highly automated system for macromolecular structure determination that can rapidly arrive at an initial partial model of a structure without significant human intervention, given moderate resolution and good quality data. This achievement has been made possible by the development of new algorithms for structure determination, maximum-likelihood molecular replacement (PHASER), heavy-atom search (HySS), template and pattern-based automated model-building (RESOLVE), automated macromolecular refinement (phenix.refine), and iterative model-building, density modification and refinement that can operate at moderate resolution (RESOLVE, AutoBuild). These algorithms are based on a highly integrated and comprehensive set of crystallographic libraries that have been built and made available to the community. The algorithms are tightly linked and made easily accessible to users through the PHENIX Wizards and the command line.

There are also a number of tools in PHENIX for handling ligands. Automated fitting of ligands into the electron density is facilitated via the LigandFit wizard. Besides being able to fit a known ligand into a difference map, the LigandFit wizard is capable to identify ligands on the basis of the difference density only. Stereo chemical dictionaries of ligands whose chemical description is not available in the supplied monomer library for the use in restrained macromolecular refinement can be generated with the electronic ligand builder and optimization workbench (eLBOW).

PHENIX builds upon Python, the Boost.Python Library, and C++ to provide an environment for automation and scientific computing. Many of the fundamental crystallographic building blocks, such as data objects and tools for their manipulation are provided by the Computational Crystallography Toolbox (cctbx). The computational tasks which perform complex crystallographic calculations are then built on top of this. Finally, there are a number of different user interfaces available in PHENIX. In order to facilitate automated operation there is the Project Data Storage (PDS) that is used to store and track the results of calculations.

The PHENIX development team consists of members from Lawrence Berkeley Laboratory (Paul Adams's group), Los Alamos National Laboratory (Tom Terwilliger's group), Cambridge University (Randy Read's group) and Duke University (the Richardsons' group). Researchers from Texas A&M University (Tom Ioerger's and Jim Sacchettini's groups) participated in the first five years of PHENIX development.

The development of PHENIX is funded by the National Institutes of Health (General Medicine) under grant P01GM063210, and the PHENIX Industrial Consortium.

Citing PHENIX

If you use PHENIX to solve a structure please cite this publication:

Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. D. Liebschner, P. V. Afonine, M. L. Baker, G. Bunkóczi, V. B. Chen, T. I. Croll, B. Hintze, L.-W. Hung, S. Jain, A. J. McCoy, N. W. Moriarty, R. D. Oeffner, B. K. Poon, M. G. Prisant, R. J. Read, J. S. Richardson, D. C. Richardson, M. D. Sammito, O. V. Sobolev, D. H. Stockwell, T. C. Terwilliger, A. G. Urzhumtsev, L. L. Videau, C. J. Williams and P. D. Adams. Acta Cryst. (2019). D75, 861-877

Publications

A number of publications describing PHENIX can be found at:

<http://www.phenix-online.org/papers/>

Working with AlphaFold2, RoseTTAFold and other predicted models

[predicted_models_index.html](#)

You can use the predicted models from AlphaFold and other prediction software in Phenix. These models can be very helpful in structure determination because the models can be very accurate over much of their length and the models come with accuracy estimates that allow removal of poorly-predicted regions.

How to use predicted models in Phenix

Phenix has tools for carrying out AlphaFold structure prediction, processing the resulting models, and for using these models in X-ray or cryo-EM structure determination. An in-depth overview with video tutorials describes these steps in detail. You can use the Phenix chatbot to get help on using any Phenix program. Just ask it any question about using Phenix. Here is a summary of the tools in Phenix. You can use `PredictModel` to access the Phenix AlphaFold server and predict the structure of a protein chain. To process predicted models, removing low-confidence regions and converting confidence estimates to B-values, you can use `ProcessPredictedModel`. The `PredictAndBuild` tool will allow you to incorporate predicted models in the structure solution workflow. `PredictAndBuild` generates predicted models and uses them to solve an X-ray structure by MR or to interpret a cryo-EM map. The tool then carries out iterative model rebuilding and prediction to improve the models. The iterative procedure allows creating more accurate predicted models than can be obtained with a simple prediction. `PredictAndBuild` can automatically generate a fairly accurate model starting from just a sequence file and either cryo-EM half-maps or X-ray data. Additionally, it provides morphed versions of unrefined predicted models that can be useful as reference models for refinement. Phenix resources for using predicted models: Overview and video tutorials: AlphaFold and Phenix Predict a structure with AlphaFold Processing a predicted model Solve an X-ray or cryo-EM structure with `PredictAndBuild` Docking a processed predicted model in a cryo-EM map Processing, docking and rebuilding a predicted model in a cryo-EM map Trimming overlapping parts of models

Here is a summary of the tools in Phenix.

You can use `PredictModel` to access the Phenix AlphaFold server and predict the structure of a protein chain.

To process predicted models, removing low-confidence regions and converting confidence estimates to B-values, you can use `ProcessPredictedModel`.

The `PredictAndBuild` tool will allow you to incorporate predicted models in the structure solution workflow. `PredictAndBuild` generates predicted models and uses them to solve an X-ray structure by MR or to interpret a cryo-EM map. The tool then carries out iterative model rebuilding and prediction to improve the models. The iterative procedure allows creating more accurate predicted models than can be obtained with a simple prediction. `PredictAndBuild` can automatically generate a fairly accurate model starting from just a sequence file and either cryo-EM half-maps or X-ray data. Additionally, it provides morphed versions of unrefined predicted models that can be useful as reference models for refinement.

Phenix resources for using predicted models:

- Overview and video tutorials: AlphaFold and Phenix
- Predict a structure with AlphaFold
- Processing a predicted model

- Solve an X-ray or cryo-EM structure with PredictAndBuild
- Docking a processed predicted model in a cryo-EM map
- Processing, docking and rebuilding a predicted model in a cryo-EM map
- Trimming overlapping parts of models

Crystallographic Structure Deposition

xray-structure-deposition.html

Background

Once a crystallographic structure is solved, it is usually deposited in the Worldwide Protein Data Bank (wwPDB). This is necessary for publication as most journals require a PDB ID to be included in any manuscript that describes a new crystallographic structure. The wwPDB is an invaluable resource that contains over 190,000 structures (as of July 2021).

Procedure

The model files (PDB and mmCIF) generated by phenix.refine contain the information required when depositing a structure to the wwPDB. However, the wwPDB currently only accepts the mmCIF (i.e., PDBx/mmCIF) model files for deposition.

To generate a model file suitable for deposition, we recommend a two-stage process:

- By default, phenix.refine will output model files in mmCIF and PDB format. If you turned off the option for mmCIF output, run a final cycle of phenix.refine that writes mmCIF files for model and data. This can be set in the Output section of the GUI.
- You can then process the model file (mmCIF) and a sequence file with mmtbx.prepare_pdb_deposition to create a mmCIF file with the sequence. This program requires the full sequence for the macromolecule to be provided. In the GUI, this program is in the "PDB Deposition" section of tools.

How to use *mmtbx.prepare_pdb_deposition*: [Click here](#)

Related programs

- phenix.get_pdb_validation_report: This program submits your model and data in CIF format to the PDB OneDep server to get a validation report in PDF and XML formats.
- phenix.table_one: This program generates the table of crystallographic statistics required by most scientific journals.

References

Announcing mandatory submission of PDBx/mmCIF format files for crystallographic depositions to the Protein Data Bank (PDB). P.D. Adams, P.V. Afonine, K. Baskaran, H.M. Berman, J. Berrisford, G. Bricogne, D.G. Brown, S.K. Burley, M. Chen, Z. Feng, C. Flensburg, A. Gutmanas, J.C. Hoch, Y. Ikegawa, Y. Kengaku, E. Krissinel, G. Kurisu, Y. Liang, D. Liebschner, L. Mak, J.L. Markley, N.W. Moriarty, G.N. Murshudov, M. Noble, E. Peisach, I. Persikova, B.K. Poon, O.V. Sobolev, E.L. Ulrich, S. Velankar, C. Vonnrhein, J. Westboork, M. Wojdyr, M. Yokochi, and J.Y. Young. Acta Cryst. D75, 451-454 (2019).

- Announcing mandatory submission of PDBx/mmCIF format files for crystallographic depositions to the Protein Data Bank (PDB). P.D. Adams, P.V. Afonine, K. Baskaran, H.M. Berman, J. Berrisford, G. Bricogne, D.G. Brown, S.K. Burley, M. Chen, Z. Feng, C. Flensburg, A. Gutmanas, J.C. Hoch, Y. Ikegawa, Y. Kengaku, E. Krissinel, G. Kurisu, Y. Liang, D. Liebschner, L. Mak, J.L. Markley, N.W. Moriarty, G.N. Murshudov, M. Noble, E. Peisach, I. Persikova, B.K. Poon, O.V. Sobolev, E.L. Ulrich, S. Velankar, C. Vonnrhein, J. Westboork, M. Wojdyr, M. Yokochi, and J.Y. Young. Acta Cryst. D75, 451-454 (2019).

Crystallographic Structure Solution with Phenix

[xray_index.html](#)

X-ray Crystallography

Click on the boxes in the figure to read an overview of each step. Also see the AlphaFold models in Phenix overview and video tutorials for using AlphaFold models in Phenix. You can use the Phenix chatbot to get help on using any Phenix program. Just ask it any question about using Phenix. GUI documentation for most used programs: Predict a structure with AlphaFold using PredictModel Automated structure determination using AlphaFold models with PredictAndBuild Check data quality with Xtrriage Experimental phasing with AutoSol Molecular replacement with Phaser Model building with AutoBuild Refine your model with phenix.refine Validation tools in Phenix Fit your ligand with LigandFit How to generate geometry restraints Prepare your model and data for deposition Complete list of programs for each structure solution step Neutron Crystallography Phenix supports neutron scattering lengths and features joint X-N refinement. Microcrystal Electron Diffraction (MicroED) Phenix supports electron scattering factors for neutral atoms.

GUI documentation for most used programs:

- Predict a structure with AlphaFold using PredictModel
- Automated structure determination using AlphaFold models with PredictAndBuild
- Check data quality with Xtrriage
- Experimental phasing with AutoSol
- Molecular replacement with Phaser
- Model building with AutoBuild
- Refine your model with phenix.refine
- Validation tools in Phenix
- Fit your ligand with LigandFit
- How to generate geometry restraints
- Prepare your model and data for deposition
- Complete list of programs for each structure solution step

Neutron Crystallography

Phenix supports neutron scattering lengths and features joint X-N refinement.

Microcrystal Electron Diffraction (MicroED)

Phenix supports electron scattering factors for neutral atoms.

AQuaRef: AI-enabled Quantum Refinement of macromolecular models

AQuaRef.html

Authors

Roman Zubatyuk, Malgorzata Biczysko, Kavindri Ranasinghe, Nigel W. Moriarty, Hatice Gokcan, Holger Kruse, Billy K. Poon, Paul D. Adams, Mark P. Waller, Adrian E. Roitberg, Olexandr Isayev, Pavel V. Afonine

Purpose

The purpose of AQuaRef is to refine atomic model coordinates using geometric restraints derived from AI-accelerated quantum-mechanical calculations. Real- and reciprocal-space refinements are supported using crystallographic or cryo-EM experimental data.

Expectations

Atomic models resulting from AQuaRef refinement are expected to have improved geometry, as evaluated by standard validation tools such as MolProbity, Rama-Z, and hydrogen bond skew-kurtosis plots. They should also maintain a similar or improved fit to the experimental data compared to standard classic refinement.

Requirements and Current Limitations

Protein-only (plus water) models are supported at the moment. Static disorder (i.e., alternative conformations) is not supported, except for exchangeable H/D sites in case of refinement against neutron diffraction data.

If a model contains unsupported entity (e.g., a metal ion) or alternative conformations, it is still possible to refine it but unsupported entities need to be excluded using 'exclude' keyword, as shown below in Examples.

The atomic model needs to be nearly final (in the later stages of standard classic refinement) and atom-complete (e.g., all hydrogen atoms are present explicitly, no missing side chains). The model is also expected to have meaningful geometry, such as being devoid of overlapping (clashing) atoms or severely distorted covalent bonds.

Hardware: While this can possibly work on both CPU and GPU computers, currently available builds are specific to Linux machines with CUDA 11 or 12. Additionally, runtimes on CPU-based computers are expected to be much slower compared to GPUs.

No GUI: Command line only.

Availability

AQuaRef is available as part of Phenix or separately from Phenix as part of Q|R refinement project (www.qrefine.com or <https://github.com/qrefine/qrefine>).

Examples

A self-consistent, quick to run, simple example is distributed with the software (Phenix, Q|R) and located in `phenix/modules/qrefine/examples/AQuaRef` folder. Check README file for details. If using AQuaRef for the first time, we suggest to run this example first in order to make sure all works as expected.

Reciprocal space refinement:

```
qr.aquaref model.pdb data.mtz
```

Real space refinement:

```
qr.aquaref model.pdb map.mrc
```

Optimization of model geometry (no experiemntal data used):

```
qr.aquaref model.pdb
```

Reciprocal space refinement excluding Mg and ligand ATP:

```
qr.aquaref model.pdb data.mtz exclude="element MG or name ATP"
```

Literature

AQuaRef: Machine learning accelerated quantum refinement of protein structures

<https://doi.org/10.1101/2024.07.21.604493>

MRage: automated molecular replacement

MRage.html

Contents

- Input
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MRage is generic molecular replacement program that is designed to run efficiently with a large number of possible models on parallel hardware. It accepts multiple model definitions that are internally processed with integrated tools to search models and used for molecular replacement. Identified solutions can be used to speed up the search even further by model assembly algorithms and limiting search space.

Input

The composition of the unit cell is specified by inputting components, and potential models for the corresponding component in a hierarchical way. It is recommended to specify the sequence of a component, which allows for better handling of partial models (see Composition stage). A stoichiometry of each component in the final assembly is also assigned at this stage. The overall count of assemblies in the unit cell can either be input or determined automatically from the Matthews coefficient. In case no model is known for a component, the list of potential models should be left empty.

Models

Models for a component can be specified stepwise with the amount of preprocessing required. The following options are available:

ensemble. This model is used without modification. model collection. These models will be superposed and optionally trimmed using phenix.ensembl, but no modification of sequence, sidechains, or B-factors will be

performed. template. This model is run through phenix.sculptor with a given set of active protocols (default: use all protocols) to remove unrelated chain segments and solvent, trim sidechains and weight the model according to structure and homology. An alignment can be specified to be used to process the structure; in case this is omitted, phenix.muscle will be used to align the model to the target sequence. homology. Homology search results. Hits will be fetched from the wwPDB, and preprocessed as a template. It is possible to limit the homology search to the top N hits. search. Performs a BLAST search for homologs and processes the results as described above. Requires component sequence.

- ensemble. This model is used without modification.
- model collection. These models will be superposed and optionally trimmed using phenix.enssembler, but no modification of sequence, sidechains, or B-factors will be performed.
- template. This model is run through phenix.sculptor with a given set of active protocols (default: use all protocols) to remove unrelated chain segments and solvent, trim sidechains and weight the model according to structure and homology. An alignment can be specified to be used to process the structure; in case this is omitted, phenix.muscle will be used to align the model to the target sequence.
- homology. Homology search results. Hits will be fetched from the wwPDB, and preprocessed as a template. It is possible to limit the homology search to the top N hits.
- search. Performs a BLAST search for homologs and processes the results as described above. Requires component sequence.

These steps are performed on-demand, i.e. if no clear solution could be found with available ensembles, the program processes the next template, or fetches the next homology search hit, etc.

Symmetry exploration

Specifies the set of space groups to try in the search. Currently, there are three options available:

dataset. The space group as specified on the input or read from the mtz-file. enantiomorph. The space group has been determined from systematic absences. The input space group and its enantiomorph (if exists) will be used in the search. pointgroup. Only the point group of the data has been established. All space groups compatible with the point group will be tried.

- dataset. The space group as specified on the input or read from the mtz-file.
- enantiomorph. The space group has been determined from systematic absences. The input space group and its enantiomorph (if exists) will be used in the search.
- pointgroup. Only the point group of the data has been established. All space groups compatible with the point group will be tried.

Space group exploration is exhaustive and can take several macrocycles to complete. Space groups with clearly inferior results to others are incrementally removed from the search until the correct space group is established.

Search organization

MRage employs a sequential search algorithm, in which molecules are found consecutively. The algorithm consists of several stages, and also allows the exploration of potential space groups.

Composition stage

The composition of each partial structure (located in previous macrocycles) is compared with the composition of the unit cell. Available models that fit in the missing composition are selected for search. In case the modelled sequence for a search model is known, the program can account for the fact that a model may only be a partial model for a component (e.g. one domain only).

Search stage

Calculations are organized into a search tree that is explored according to depth-first traversal. For the underlying calculations, available functions from phaser [PHASER] are used.

For each partial structure and applicable model, a rotation function is calculated. This is then followed up by a translation function calculation. Each translation function peaks are checked for packing clashes, and if accepted, checked whether are significant, probable solutions or not. Calculations are performed in the order of peak score (RF/TF), model quality (sequence identity) or partial structure quality (LLG score). Exploration continues until a significant solution is found (quick mode) or all possibilities has been exhausted (full mode). There is also no thresholding step for rotation (rationale: weak signal in rotation function) and translation function peaks (rationale: packing check fast). As a consequence, more work is performed, but simultaneously this allows weighting translation function results with packing overlap, and also avoids pipe stalls, which would adversely effect scaling in a parallel environment.

In quick mode, if a significant solution is found, conventional search terminates. However, all models that are alternatives of the model that gave this clear solution are superposed onto the solution and refined. This allows the quick evaluation of model quality for a potentially large number of alternative models. Currently, SSM [SSM] is used for superposition.

Parallelisation is done at a function level. Calculations are dispatched until the number of assigned CPUs are filled up. Execution can either take place in a shared memory environment using threads and processes or using a submission queue (currently SGE, LSF and PBS are supported). The degree of parallelisation scales with the complexity of the search, and given unlimited resources, reduces the runtime to a constant. However, in simple cases (e.g. a good model that gives a clear rotation and translation peak), no parallel resources can be used.

Refinement

Peaks above a certain threshold of the best peak and those that are identified as significant solutions are subjected to refinement. After refinement is completed, a second thresholding step is performed. Top peaks will be subjected to post-processing and also carried forward for the next macrocycle. Significant peaks that are below the threshold (e.g. a correct solution for a small domain) are also input to the post-processing step, but will not be propagated to the next cycle.

Post-processing

Significant solutions are analysed for specific features so that the search could be improved.

Amalgamation. Assembly of solutions that come from the same starting point. In case the "starting point" is empty, allowed origin shifts are added to the search. In case the origin is undefined and the space group has floating origin, only the rotation component is extracted. Rotation peak salvage. Propagates significant rotation peaks to the next stage. Regular assembly recognition. Identifies local point group symmetry from the solution. Assembly completion. In case assembly definitions are available (e.g. from regular assembly completion), and a partial assembly has been located, missing molecules are filled in.

- Amalgamation. Assembly of solutions that come from the same starting point. In case the "starting point" is empty, allowed origin shifts are added to the search. In case the origin is undefined and the space group has floating origin, only the rotation component is extracted.
- Rotation peak salvage. Propagates significant rotation peaks to the next stage.
- Regular assembly recognition. Identifies local point group symmetry from the solution.
- Assembly completion. In case assembly definitions are available (e.g. from regular assembly completion), and a partial assembly has been located, missing molecules are filled in.

No actual calculation is performed, only the possibilities are noted. In case the given solution enters the next Search stage (as determined by Composition stage), these possibilities will be scored as a priority.

Space group exploration

Search progress in possible space groups is compared. Space groups having significant solutions or above the threshold (determined by the best score over all space groups) are propagated to the next macrocycle.

Output

The program logs all operations it performs. However, this may be difficult to follow on the screen, especially when multiple space groups are explored, and therefore, after each macrocycle, a summary of all searches performed is printed.

Models generated during the solution process are also saved. This includes structure files fetched from the PDB, but also alignments and homology searches performed.

Solutions

If the search finishes, solutions are written out in all likely space groups. For each space group, the top solution is selected, and all partial solutions above a threshold relative to the top solution is considered. Solutions that are fully "contained" in a higher scoring solution (i.e. all models correspond to a model in the higher scoring solution) are discarded. This way, the top solution is output even if the composition is overestimated, but at the same time removing incomplete solutions (e.g. 14 molecules out of the total 15) that cannot be distinguished based on the final score.

Solutions are written out in an internal compressed format, which is capable of storing a high number of solutions without onerous storage requirements, and can be manipulated with provided tools.

Usage

Graphical interface

As with the generic Phaser GUI, two interfaces are available for running MRage, one designed for single-component searches with minimal customization of input models, and an advanced interface that allows all available features to be used. These are launched from a single button in the main GUI, which will initially pop up an information dialog describing the available options.

The simple GUI is centered around a single input control allowing any combination of files to be dragged in:

Reflections (mandatory): these can be in any file or data format; only amplitudes or intensities are required.
 Sequence: should contain a single polymer sequence, in any format. This is highly recommended, and required if you want to perform an automatic homology search. Template models: unprocessed PDB files,

which will be treated by Sculptor as described above. Homology search: HHPred or BLAST results.

- Reflections (mandatory): these can be in any file or data format; only amplitudes or intensities are required.
- Sequence: should contain a single polymer sequence, in any format. This is highly recommended, and required if you want to perform an automatic homology search.
- Template models: unprocessed PDB files, which will be treated by Sculptor as described above.
- Homology search: HHPred or BLAST results.

In the simplest cases, only the data and sequence are required; you should then select a homology search method in the Options section. "NCBI BLAST" requires internet access; "Local BLAST" requires that BLAST is installed and available on the command line. At this point you can run the program without further input.

Other settings:

Max. number of hits to use controls how many PDB entries from the homology search are processed by the pipeline. Increasing this provides greater coverage, but at the cost of significantly longer runtimes. Resolution cutoff determines the resolution used for running Phaser; this does not need to be changed, but increasing the cutoff will result in a faster but less accurate run. Parallelization method controls how the jobs are distributed across the available computing resources. multiprocessing and threading are essentially equivalent (although in practice they may behave slightly differently), and run locally. The other options require that you have access to one of the supported queuing systems and can submit jobs from the computer where you are running the GUI. Number of CPUs controls how many jobs are run at once. It should not exceed the number of processors available, either on the local system or the queuing system. (Note: if you plan to use your computer for other tasks while MRage is running, you may wish to leave one CPU core free if possible.) Search mode: sets the thoroughness of the search, as describe in the Search stage section. Symmetry exploration determines the space groups to try (Symmetry exploration) Overall count is the number of copies of the component to search for in the asymmetric unit.

- Max. number of hits to use controls how many PDB entries from the homology search are processed by the pipeline. Increasing this provides greater coverage, but at the cost of significantly longer runtimes.
- Resolution cutoff determines the resolution used for running Phaser; this does not need to be changed, but increasing the cutoff will result in a faster but less accurate run.
- Parallelization method controls how the jobs are distributed across the available computing resources. multiprocessing and threading are essentially equivalent (although in practice they may behave slightly differently), and run locally. The other options require that you have access to one of the supported queuing systems and can submit jobs from the computer where you are running the GUI.
- Number of CPUs controls how many jobs are run at once. It should not exceed the number of processors available, either on the local system or the queuing system. (Note: if you plan to use your computer for other tasks while MRage is running, you may wish to leave one CPU core free if possible.)
- Search mode: sets the thoroughness of the search, as describe in the Search stage section.
- Symmetry exploration determines the space groups to try (Symmetry exploration)
- Overall count is the number of copies of the component to search for in the asymmetric unit.

Advanced GUI

This version of the GUI moves the description of the crystal contents and search information to a separate tab, with input options grouped by component. The first tab is essentially similar to the simple GUI, but the only input field is for the reflections file.

The composition tab contains another set of tabs for each unique component. These allow the full range of input files to be specified; some of these have additional parameters (such as the mandatory sequence identity for ensembles and model collections). The "Basic info" tab allows the NCBI BLAST search to be run interactively if desired (the results will be loaded as a homology result). A link to the HHPred server, which does not allow automated queries, is available on the "Homology searches" tab.

You must specify the component stoichiometry, i.e. the number of copies of the component in the overall assembly. This is distinct from the "Overall count" parameter in the main tab: for instance, if you are trying to solve a structure of a hexamer bound to a monomer and expect two copies of the complex in the asymmetric unit of the crystal, the overall count is two, and the stoichiometry of the components is 6 and 1 for the hexamer and monomer respectively.

Command line

phaser.MRage [command-line switches] [PHIL-format parameter files]

Command-line switches:

```
-h, --help show this help message and exit
--show-defaults print PHIL and exit
-i, --stdin read PHIL from stdin as well
-v, --verbosity set verbosity level (DEBUG, INFO, WARNING, VERBOSE)
```

PHIL arguments

Everything not starting with a dash('-') is interpreted as a PHIL argument. This can be a PHIL-format file containing parameters or a command-line assignment.

References

- Phaser crystallographic software. A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, and R.J. Read. *J Appl Crystallogr* 40, 658-674 (2007).
- Secondary-structure matching (SSM), a new tool for fast protein structure alignment in three dimensions. E. Krissinel, and K. Henrick. *Acta Crystallogr D Biol Crystallogr* 60, 2256-68 (2004).

List of all keywords

hklin = None labin = None resolution_cutoff = 2.5 no_tncs = False mode = *quick full symmetry_exploration = *dataset enantiomorph pointgroup packing_pool = 20 rotation_peaks_cutoff = 0.75 post_refinement_cutoff = 0.75 final_selection_cutoff = 0.75 b_factor_refinement = True significant_peak_threshold = 7.0 sculptor_protocols = 1 2 3 4 5 6 7 8 9 10 11 *12 13 all minimal template_equivalence = True assembly_acceptance_policy = *always observed never assembly_ensemble_creation_policy = always *observed never pdb_mirror = *rcsb pdbe pdbj exclude_pdb_ids = None Developer option to prevent specific structures from being used as search models, in order to test performance with lower sequence identity structures.crystal_symmetryunit_cell = None space_group = None outputroot = "mrage" max_solutions_to_write = 0 gui_base_dir = None Base directory in which the PHENIX GUI will create a folder containing the results. Ignored when running the program from the command line.save_aniso_data = False job_title = None Job title in PHENIX GUI, not used on command linecompositioncount = None Overall number of copies of the complete molecule (which may be multiple components). If left blank, this will be guessed based on the expected solvent content.componentsequence = None mtype = *protein dna rna Component typestoichiometry = 1 Number of copies of this component relative to the overall macromolecule. This is separate from the overall count parameter in the Basic Options tab, which defines the number of copies of the

complete unit. For instance, if you expect the basic molecule to be two copies of this component and one copy of a second component, the stoichiometry is 2. If you are searching for an unknown number of copies of a single component, you should leave the stoichiometry set to 1.

template_list = None
ensemblealignment = None
Sequence modelled by ensemblecoordinatespdb = None
identity = None
error_translation = *default
rmsd floored_chothia_and_lesk Identity to RMS conversionmodel_collectiontrim = False
coordinatespdb = None
identity = None
error_translation = *default
rmsd floored_chothia_and_lesk Identity to RMS conversiontemplatepdb = None
alignment = None
homologyfile_name = None
max_hits = 3
searchservices = ncbi local
Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
max_hits = 3
assemblyDefine known assemblyuse_assembled = False
Use assembly as model in search or only for completionlocal_searchDefine local search extentsweep = 5
Rotational search radiusextent = 5
Translational search extentcomponentDefine a component of the assemblyfile_name = None
Assembly component structureoperationModel transformation to participate in the assemblyeuler = 0 0 0
Rotationdisplacement = 0 0 0
Translationqueueutechnology = *threading
lsf sge pbs multiprocessing cpus = 1
submission_command = None
qslot_cpus = 1
simple_runParameters for simplified GUI - not intended for command-line use.
enable = False
GUI parameter - not intended for command-line use.
component_sequence = None
template_model = None
alignment = None
blast_services = ncbi local
Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
max_blast_hits = 3

- hklin = None
- labin = None
- resolution_cutoff = 2.5
- no_tncs = False
- mode = *quick full
- symmetry_exploration = *dataset enantiomorph pointgroup
- packing_pool = 20
- rotation_peaks_cutoff = 0.75
- post_refinement_cutoff = 0.75
- final_selection_cutoff = 0.75
- b_factor_refinement = True
- significant_peak_threshold = 7.0
- sculptor_protocols = 1 2 3 4 5 6 7 8 9 10 11 *12 13 all minimal
- template_equivalence = True
- assembly_acceptance_policy = *always observed never
- assembly_ensemble_creation_policy = always *observed never
- pdb_mirror = *rcsb pdbe pdbj
- exclude_pdb_ids = None
Developer option to prevent specific structures from being used as search models, in order to test performance with lower sequence identity structures.
- crystal_symmetryunit_cell = None space_group = None
- unit_cell = None
- space_group = None

- outputroot = "mrage" max_solutions_to_write = 0 gui_base_dir = None Base directory in which the PHENIX GUI will create a folder containing the results. Ignored when running the program from the command line. save_aniso_data = False job_title = None Job title in PHENIX GUI, not used on command line
- root = "mrage"
- max_solutions_to_write = 0
- gui_base_dir = None Base directory in which the PHENIX GUI will create a folder containing the results. Ignored when running the program from the command line.
- save_aniso_data = False
- job_title = None Job title in PHENIX GUI, not used on command line
- compositioncount = None Overall number of copies of the complete molecule (which may be multiple components). If left blank, this will be guessed based on the expected solvent content. componentsequence = None mtype = *protein dna rna Component type stoichiometry = 1 Number of copies of this component relative to the overall macromolecule. This is separate from the overall count parameter in the Basic Options tab, which defines the number of copies of the complete unit. For instance, if you expect the basic molecule to be two copies of this component and one copy of a second component, the stoichiometry is 2. If you are searching for an unknown number of copies of a single component, you should leave the stoichiometry set to 1. template_list = None ensemblealignment = None Sequence modelled by ensemble coordinatespdb = None identity = None error_translation = *default rmsd floored_chothia_and_lesk Identity to RMS conversion model_collectiontrim = False coordinatespdb = None identity = None error_translation = *default rmsd floored_chothia_and_lesk Identity to RMS conversion templatepdb = None alignment = None homologyfile_name = None max_hits = 3 searchservices = ncbi local Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access). max_hits = 3
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- sequence = None
- mtype = *protein dna rna Component type
- stoichiometry = 1 Number of copies of this component relative to the overall macromolecule. This is separate from the overall count parameter in the Basic Options tab, which defines the number of copies of the complete unit. For instance, if you expect the basic molecule to be two copies of this component

and one copy of a second component, the stoichiometry is 2. If you are searching for an unknown number of copies of a single component, you should leave the stoichiometry set to 1.

- `template_list = None`
- `ensemblealignment = None` Sequence modelled by ensemble
`coordinatespdb = None` identity = None
`error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `alignment = None` Sequence modelled by ensemble
- `coordinatespdb = None` identity = None
`error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `pdb = None`
- `identity = None`
- `error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `model_collectiontrim = False` `coordinatespdb = None` identity = None
`error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `trim = False`
- `coordinatespdb = None` identity = None
`error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `pdb = None`
- `identity = None`
- `error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `templatepdb = None` `alignment = None`
- `pdb = None`
- `alignment = None`
- `homologyfile_name = None` `max_hits = 3`
- `file_name = None`
- `max_hits = 3`
- `searchservices = ncbi local` Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
`max_hits = 3`
- `services = ncbi local` Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
- `max_hits = 3`
- `assemblyDefine` known assembly
`use_assembled = False` Use assembly as model in search or only for completion
`local_searchDefine` local search extents
`sweep = 5` Rotational search radius
`extent = 5` Translational search extent
`componentDefine` a component of the assembly
`file_name = None` Assembly component structure
`operationModel` transformation to participate in the assembly
`euler = 0 0 0` Rotation
`displacement = 0 0 0` Translation
- `use_assembled = False` Use assembly as model in search or only for completion
- `local_searchDefine` local search extents
`sweep = 5` Rotational search radius
`extent = 5` Translational search extent
- `sweep = 5` Rotational search radius

- extent = 5 Translational search extent
- componentDefine a component of the assemblyfile_name = None Assembly component structureoperationModel transformation to participate in the assemblyeuler = 0 0 0 Rotationdisplacement = 0 0 0 Translation
- file_name = None Assembly component structure
- operationModel transformation to participate in the assemblyeuler = 0 0 0 Rotationdisplacement = 0 0 0 Translation
- euler = 0 0 0 Rotation
- displacement = 0 0 0 Translation
- queueutechnology = *threading lsf sge pbs multiprocessing cpus = 1 submission_command = None qslot_cpus = 1
- technology = *threading lsf sge pbs multiprocessing
- cpus = 1
- submission_command = None
- qslot_cpus = 1
- simple_runParameters for simplified GUI - not intended for command-line use.enable = False GUI parameter - not intended for command-line use.component_sequence = None template_model = None alignment = None blast_services = ncbi local Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).max_blast_hits = 3
- enable = False GUI parameter - not intended for command-line use.
- component_sequence = None
- template_model = None
- alignment = None
- blast_services = ncbi local Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
- max_blast_hits = 3

Running phenix.refine using AFITT gradients for ligands

afitt.html

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- Literature

Authors

Pawel Janowski and Nigel W. Moriarty

Theory and Background

AFITT refinement in Phenix combines the power and functionality of Phenix crystallographic refinement with OpenEye AFITT's [Wlodek2006] implementation of the MMFF94s [Halgren1996] force field for accurate modeling of small molecule stereochemistry. In combination they provide the user with a cutting-edge crystallographic refinement package while ensuring chemically sensible ligand geometry that significantly reduces conformational strain as compared to traditional refinement. Phenix.refine is a component of the highly regarded Phenix suite of crystallography software. AFITT is OpenEye's comprehensive package for ligands in crystallography. It provides an automated real space ligand fitting process, refinement dictionary generator, implementation of MMFF94, MMFF94s, PM3 and AM1 force fields and the OpenEye's core SZYBKI optimizer.

Phenix.refine employs a maximum likelihood approach to minimize the following residual or energy function:

$$E_{\text{Phenix}} = E_{\text{x-ray}} + E_{\text{geometry}}$$

where, in the presence of ligands, the second term in the above equation can be further broken down into:

$$E_{\text{geometry}} = E_{\text{protein}} + E_{\text{ligand_non-bonded}} + E_{\text{ligand_bonded}}$$

Phenix-AFITT replace Phenix's bonded ligand term with an AFITT term, thus yielding the following final energy function:

$$E_{\text{Phenix-AFITT}} = E_{\text{x-ray}} + E_{\text{geometry}} - E_{\text{ligand_bonded}}^{\text{Phenix}} + E_{\text{ligand_bonded}}^{\text{AFITT}}$$

Phenix's implementation of AFITT is versatile, easy to use and powerful. It can support any ligand for which a cif dictionary has been created. Refinements can include different types of ligands and multiple instances of each ligand type. Support for ligands with full or partial alternate conformations is fully integrated as is refinement of ligands covalently bound to the macromolecule.

Running refinement with Phenix and AFITT

First, the user should make sure that the latest version of Phenix and AFITT are installed. On Linux and OSX, make sure that the OE_EXE environment variable is set and points to the executable (usually flynn). For example:

```
[~:~]$ echo $OE_EXE
```

```
/usr/local/bin/flynn
```

Second prepare the .pdb, .mtz and .cif dictionaries for the ligand in your model. Now running Phenix-AFITT refinement is just a question of running standard Phenix.refine with some additional keywords. For example like this:

```
phenix.refine mymodel.pdb mymodel.mtz myligand.cif use_afitt=True  
afitt.ligand_file=myligand.cif afitt.ligand_names=UNL,BCL
```

The above command will run Phenix.refine with AFITT energy and gradients applied to the ligands UNL and BCL according to the topology provided in the file myligand.cif. The following is the full list of AFITT-related keyword that the user can specify on the command line when running Phenix.refine:

REQUIRED KEYWORDS use_afitt - turn on ligand refinement with AFITT. (default = False)

afitt.ligand_file_name - relative path to cif dictionary file that defines the ligand topology for AFITT. Note that this can be different from the cif file provided to Phenix.refine on the command line without any keyword. The former specifies the topology of the ligand for AFITT. The actual bond and angle parameters in it are irrelevant. The latter specifies the both topology and parameters of non-standard residues for Phenix. Any ligands that are not refined by AFITT (using the "ligand_names" keyword) will be refined by Phenix.refined using the parameters in this file. afitt.ligand_names - three letter ligand name codes. If multiple ligand types are being refined, separate each three letter code by a comma, no spaces. For a given ligand name, AFITT will refine all instances of that ligand as well as alternate conformations of each ligand. OPTIONAL KEYWORDS afitt.ff - AFITT force field to use. Options are: mmff94s, mmff94, am1, pm3. (default=mmff94s) afitt.scale - weight to place on the AFITT ligand energy term. (default=10)

REQUIRED KEYWORDS

use_afitt - turn on ligand refinement with AFITT. (default = False) afitt.ligand_file_name - relative path to cif dictionary file that defines the ligand topology for AFITT. Note that this can be different from the cif file provided to Phenix.refine on the command line without any keyword. The former specifies the topology of the ligand for AFITT. The actual bond and angle parameters in it are irrelevant. The latter specifies the both topology and parameters of non-standard residues for Phenix. Any ligands that are not refined by AFITT (using the "ligand_names" keyword) will be refined by Phenix.refined using the parameters in this file. afitt.ligand_names - three letter ligand name codes. If multiple ligand types are being refined, separate each three letter code by a comma, no spaces. For a given ligand name, AFITT will refine all instances of that ligand as well as alternate conformations of each ligand.

use_afitt - turn on ligand refinement with AFITT. (default = False)

afitt.ligand_file_name - relative path to cif dictionary file that

defines the ligand topology for AFITT. Note that this can be different from the cif file provided to Phenix.refine on the command line without any keyword. The former specifies the topology of the ligand for AFITT. The actual bond and angle parameters in it are irrelevant. The latter specifies the both topology and parameters of non-standard residues for Phenix. Any ligands that are not refined by AFITT (using the "ligand_names" keyword) will be refined by Phenix.refined using the parameters in this file.

afitt.ligand_names - three letter ligand name codes. If multiple

ligand types are being refined, separate each three letter code by a comma, no spaces. For a given ligand name, AFITT will refine all instances of that ligand as well as alternate conformations of each ligand.

OPTIONAL KEYWORDS

afitt.ff - AFITT force field to use. Options are: mmff94s, mmff94, am1, pm3. (default=mmff94s) afitt.scale - weight to place on the AFITT ligand energy term. (default=10)

afitt.ff - AFITT force field to use. Options are: mmff94s, mmff94, am1, pm3. (default=mmff94s)

afitt.scale - weight to place on the AFITT ligand energy term. (default=10)

Auxiliary functions

Two auxiliary command line programs are provided for convenience. mmtbx.afitt returns the AFITT ligand conformational energy given a pdb file (for coordinates) and cif file (for topology) and the three letter names of the ligands the user is querying. Optionally the user can specify the AFITT force field to use using the "-ff" keyword (default=MMFF94s). Example usage:

```
[~/work/afitt:]$ mmtbx.afitt 1cvu.pdb 1cvu.cif BOG,ACD
BOG_702_ AFITT_ENERGY: 2152.1468
BOG_703_ AFITT_ENERGY: 2151.5215
BOG_704_ AFITT_ENERGY: 2137.6938
BOG_2702_ AFITT_ENERGY: 2145.0459
BOG_2704_ AFITT_ENERGY: 2099.0437
ACD_701_ AFITT_ENERGY: 136.1296
ACD_2701_ AFITT_ENERGY: 99.7701
```

mmtbx.afitt_fd runs a finite difference test to check that the Phenix-AFITT energy and gradients are being properly calculated. This is the first test to run if you think something may be wrong with the implementation of the refinement algorithm. mmtbx.afitt_fd requires a pdb file, cif file, ligand names and atom number on whose x-coordinate the finite difference test will be calculated. An optional "-v" keyword provides more verbose output about the individual terms of the energy function. Example usage:

```
[~/work/afitt:]$ mmtbx.afitt_fd 1cvu.pdb 1cvu.cif ACD 9321
-> 0.134306495
-> 0.134306174
TEST PASSES: (analytical - numerical)= -0.000000321
```

As always, running an auxiliary program with the "-h" keyword provides helpful information about the program.

Example ligand conformational energies after traditional refinement and after refinement with Phenix-AFITT. Energies are shown as a percentage of the energy of the ligand conformation deposited in the PDB.

Literature

AI and Phenix

ai.html

Phenix is supplied with several AI tools that can help you decide what to do and to help you solve your structure:

- The Phenix chatbot will answer your questions about Phenix and how to use Phenix tools and how to solve a structure
- The Phenix AI agent can automatically analyze your X-ray or cryo-EM data and solve your structure. It can also run many Phenix tools for you.
- The Phenix AI analysis tool can look at your log file and suggest what to do next.

If you are a beginner, ask the chatbot how to solve your structure if you have X-ray data (or cryo-EM data). Use the AI agent to automatically set up and analyze your data for you, and use the AI analysis tool and the chatbot to interpret your results.

If you are an advanced user, try the AI agent as a tool for quickly screening your data.

If you are either a beginner or an advanced user, also have a look at the overall Phenix documentation for tips and ideas for solving your structure.

Running phenix.refine using Amber gradients

amber.html

Contents

- Authors
- Theory and Background
- Running refinement with Phenix and Amber

Authors

Principal authors: Nigel W. Moriarty, Pawel Janowski, Hai Nguyen & David A. Case

Contributions from: Jason Swails (Python Sander API), Romain Wolf (pdb4amber which is used in AmberPrep)

Theory and Background

Amber refinement in Phenix combines the power and functionality of Phenix crystallographic refinement with the Amber molecular mechanics force field for geometry restraints. Amber restraints replace the conventional set of restraints. In addition to bond, angle and dihedral terms, electrostatics and van der Waals attractive/dispersive interactions are computed. Calls to Amber's energy and gradient calculation engines are seamlessly integrated through internal libraries shared with Phenix. Refinement with Amber molecular mechanics has been shown to improve model quality and reduce overfitting as compared to conventional restraints.

Running refinement with Phenix and Amber

From Phenix 1.16, Amber is distributed with the standard installer for unix and OSX. To run Amber in Phenix, in addition to a PDB file, a topology and a coordinate file must be provided for Amber. From Phenix 1.18, you can allow the phenix.refine GUI to run AmberPrep, automatically creating the files needed. You can also easily create these files by using the provided AmberPrep utility via the AmberPrep GUI or via the commandline:

```
phenix.fetch_pdb 4Lzt --mtz
phenix.AmberPrep 4Lzt.pdb
```

which will generate four files.

```
4amber_4Lzt.prmtop
4amber_4Lzt.rst7
4amber_4Lzt.order
4phenix_4Lzt.pdb
```

The first two are the Amber files and the last one is the PDB file that matches the two Amber files for atom order and must be used in phenix.refine replacing 4Lzt.pdb. Having created the input files, refinement can be run as follows:

```
phenix.refine 4phenix_4Lzt.pdb 4Lzt.mtz use_amber=True topology_file_name=4amber_4Lzt.prmtop coordinate_file_name=4phenix_4Lzt.pdb
xray_data.r_free_flags.generate=True
```

Here is list of keywords:

REQUIRED KEYWORDS `use_amber` - turn on refinement with Amber geometry restraints. ("True" or "False"; default = False) `topology_file_name` - relative path to the Amber topology file. This is the prmtop file created by AmberPrep. `coordinate_file_name` - relative path to the Amber coordinate file. This is the rst7 file created by AmberPrep. OPTIONAL KEYWORDS `wxc_factor` - relative weight factor between the x-ray and the Amber geometry restraint terms in refinement. By default this is set to 0.1 which multiplies the standard `wxc_scale` which has a default of 0.5 for the x-ray and E&H restraints. (Float; default=0.1)

REQUIRED KEYWORDS

`use_amber` - turn on refinement with Amber geometry restraints. ("True" or "False"; default = False)
`topology_file_name` - relative path to the Amber topology file. This is the prmtop file created by AmberPrep.
`coordinate_file_name` - relative path to the Amber coordinate file. This is the rst7 file created by AmberPrep.

`use_amber` - turn on refinement with Amber geometry restraints. ("True" or "False"; default = False)

`topology_file_name` - relative path to the Amber topology file.

This is the prmtop file created by AmberPrep.

`coordinate_file_name` - relative path to the Amber coordinate file.

This is the rst7 file created by AmberPrep.

OPTIONAL KEYWORDS

`wxc_factor` - relative weight factor between the x-ray and the Amber geometry restraint terms in refinement. By default this is set to 0.1 which multiplies the standard `wxc_scale` which has a default of 0.5 for the x-ray and E&H restraints. (Float; default=0.1)

`wxc_factor` - relative weight factor between the x-ray and

the Amber geometry restraint terms in refinement. By default this is set to 0.1 which multiplies the standard `wxc_scale` which has a default of 0.5 for the x-ray and E&H restraints. (Float; default=0.1)

ADDITIONAL INFORMATION:

1. We also provide the possibility to energetically optimize model coordinates using Amber's force field rather than the set of Engh & Huber restraints. This leads to models of improved quality. This minimization can be run as follows:

```
phenix.geometry_minimization 4amber_xyz.pdb amber.use=True topology_file_name=4amber_4Lzt.prmtop coordinate_file_name=4amber_4Lzt.rst7
```

2. AmberPrep contains additional options that the user may wish to use. Run:

```
phenix.AmberPrep --show-defaults
```

to view all of the options. Here is a list of the optional keywords:

`minimise` - type of energy minimization to perform. Possible options: `amber_all` - optimize coordinates of all atoms using Amber's minimizer `amber_h` - optimize only hydrogen positions using Amber's minimizer (default) `phenix_all` - optimize coordinates of all atoms using phenix minimizer `off` - do not optimize atoms `clean` - remove temporary files (True/False, default=True) `redq` - use Amber's redq force field instead of the default (currently ff14SB). redq is a version of ff14SB that mitigates the partial charges on charged amino acid side chains. In theory this should account for the electrostatic screening effects of the unmodelled solvent molecules. In practice our tests have not shown any improvement in refinement when using the reduced charge (redq) force field, but users are free to experiment (True/False, default=False)

`minimise` - type of energy minimization to perform. Possible options:

amber_all - optimize coordinates of all atoms using Amber's minimizer
amber_h - optimize only hydrogen positions using Amber's minimizer (default)
phenix_all - optimize coordinates of all atoms using phenix minimizer
off - do not optimize atoms

amber_all - optimize coordinates of all atoms using Amber's minimizer

amber_h - optimize only hydrogen positions using Amber's minimizer (default)

phenix_all - optimize coordinates of all atoms using phenix minimizer

off - do not optimize atoms

clean - remove temporary files (True/False, default=True)

redq - use Amber's redq force field instead of the default (currently

ff14SB). redq is a version of ff14SB that mitigates the partial charges on charged amino acid side chains. In theory this should account for the electrostatic screening effects of the unmodelled solvent molecules. In practice our tests have not shown any improvement in refinement when using the reduced charge (redq) force field, but users are free to experiment (True/False, default=False)

3. Refinement with Amber currently has several limitations. All atoms must be explicitly modelled and there can be no gaps in the peptide or nucleic acid backbone. If hydrogen atoms are missing, they will be added automatically by AmberPrep but missing heavy atoms should first be manually modelled in by the user. Amber does not currently model alternate conformations and AmberPrep will only retain the first alternate conformer of each atom. Solutions to these two limitations are currently in development.

4. Metal-containing ligands are not supported by Amber.

5. Atoms at special positions will most often produce high energy clashes with Amber and usually will not work. If the atom is a solvent molecule, the user may try to refine without it. If the structure contains atoms at special positions, a warning will be issued when running AmberPrep.

FOR ADVANCED USERS:

1. To use MDGX instead of Sander for gradient calculation, use md_engine=mdgx keyword for phenix.refine. The default md_engine is Sander.

2. phenix.refine with Amber also has a hidden wxc_factor parameter. In ordinary refinement the x-ray term is scaled by the ratio of the gradient norms between the x-ray and geometry terms times the wxc_factor (default 0.5 when using E&H; recommended 0.025 when using Amber). If wxc_factor is set, the ratio of the gradient norms will be ignored and the x-ray term will be scaled by wxc_factor directly. This is an experimental feature and has not been shown to improve refinement in any way.

3. The advanced user may wish to change the Amber force field used to impose geometry restraints. By default the force field used is ff14SB. This can easily be changed by modifying AmberPrep script, run_tleap() method, to source a different force field from Amber.

4. phenix.AmberPrep performs several important actions to prepare files for refinement with Phenix and Amber. This includes:

checks for existence of ligands and non-standard small molecules. Derives Amber parameters using eLBOW and Antechamber cleans and modifies PDB file for conformity with Amber runs TLEaP to prepare Amber topology and coordinate files runs Amber's XtalUtilities to prepare Amber inputs that conform to the given crystallographic space group adds missing hydrogen atoms and possibly missing side chain heavy atoms creates a PDB input file for Phenix that corresponds in atom order and content to the Amber input files

- checks for existence of ligands and non-standard small molecules. Derives Amber parameters using eLBOW and Antechamber
- cleans and modifies PDB file for conformity with Amber
- runs TLEaP to prepare Amber topology and coordinate files
- runs Amber's XtalUtilities to prepare Amber inputs that conform to the given crystallographic space group
- adds missing hydrogen atoms and possibly missing side chain heavy atoms
- creates a PDB input file for Phenix that corresponds in atom order and content to the Amber input files

Analyzing alternative conformations matching a map with `analyze_alt_conf`

`analyze_alt_conf.html`

Author(s)

- `analyze_alt_conf`: Tom Terwilliger

Purpose

The purpose of `analyze_alt_conf` is to analyze alternative conformations (focusing on side-chains) for a structure and an optional comparison structure.

The problem that it solves is that if you have multiple conformations present in two structures, identifying which ones match and which ones are unique is tedious.

How `analyze_alt_conf` works:

System Message: WARNING/2 (<string>, line 21)

Title underline too short.

```
How analyze_alt_conf works:
-----
```

This tool first superposes all models that are supplied. Then it analyzes the unique residue conformations in each model that is supplied. Residues that superpose with an rmsd of `min_rotamer_rmsd` (default of 0.25 Å) or better are considered the same conformer; those with a larger rmsd are considered different.

If there are two models supplied, the tool then compares the unique conformers in the first and second models. It identifies which altlocs in the first model (m1) match which altlocs in the second model (m2), and which are unique in one model or the other.

Examples

Standard run of `analyze_alt_conf`:

System Message: WARNING/2 (<string>, line 38)

Title underline too short.

```
Standard run of analyze_alt_conf:
-----
```

Running `analyze_alt_conf` is easy. From the command-line you can type:

```
phenix.analyze_alt_conf model.pdb other_model.pdb
```

where `model.pdb` is the model you would like to use as a starting point and `other_model.pdb` is an optional comparison model.

Possible Problems

The procedure only analyzes alternative side-chain conformers, not main-chain. If you main chain has substantial alternative conformations (not just slight adjustments to match the side chain conformers), you will need to use another approach.

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesmodel = None Input model
analyze_alt_confmin_rotamer_rmsd = 0.25 Minimum rmsd between residues. If atoms in residue differ by less than this, consider them the same.
match_any_atom = True Ignore atom names when comparing residues. Just take closest pairs between structures
guiGUI-specific parameter required for output directory
output_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmodel = None Input model
- model = None Input model
- analyze_alt_confmin_rotamer_rmsd = 0.25 Minimum rmsd between residues. If atoms in residue differ by less than this, consider them the same.
- match_any_atom = True Ignore atom names when comparing residues. Just take closest pairs between structures
- min_rotamer_rmsd = 0.25 Minimum rmsd between residues. If atoms in residue differ by less than this, consider them the same.
- guiGUI-specific parameter required for output directory
- output_dir = None

AI analysis of Phenix results

analyze_log.html

Authors

- Nigel Moriarty, Peter Zwart, Thomas C Terwilliger

Purpose

The Phenix AI analysis tool is designed to help you interpret your output files from a run in the Phenix GUI. It analyzes your log file and any output written to the GUI and the names of any output files that are supplied to the GUI and creates a summary of the run. Then it analyzes this summary in the context of the Phenix documentation to describe how this run fits into the framework of structure determination and suggests next steps. The AI analysis tool can be accessed in the Phenix GUI in the Results section of most Phenix GUI runs, next to the Log file button. You can also access it under the Tools menu.

Google and Cohere API Keys

To run the AI analysis, you will need Google and Cohere API keys. The Google key will let you run Gemini from the Phenix server, and the Cohere key will let you run Cohere Langchain tools. Cohere keys are free. Google API keys are free (up to \$300 in usage) for up to 3 months. After that you will be charged, but you can see from your usage in your free trial period how much (or little) you would be charged. You can get the keys from the AI section of the Preferences menu in the Phenix GUI.

You can run a AI analyses without getting your own key as a shared set of keys is supplied, but you should get your own set once you get going so that others can use the shared keys.

How the AI analysis works

The AI analysis is carried out in two steps. In the first step, the log file, along with text written to the GUI and names of output files supplied to the GUI are summarized using a Cohere Langchain analysis. In the second step, the summary from the first step is analyzed in the context of all the Phenix documentation, transcripts of Phenix tutorial videos, and Phenix publications to provide background and to suggest next steps to take.

This type of AI does not save or learn from your questions. However the information in your log file is sent to the Phenix server, and from there, on to Cohere and Google gemini.

What to do with the AI analysis

The purpose of the AI analysis is to help you interpret your output and to suggest ideas for next steps. You should keep in mind that this tools can make mistakes and give you incorrect interpretations and poor suggestions at times, so you want to just take the information as suggestions to think about.

You can combine this AI analysis with the Phenix chatbot . The chatbot can give you interactive answers to your questions using the same database of information as the AI analysis. This allows you to follow up on the AI analysis with questions to the chatbot. You can also paste part of the output from the AI analysis into the chatbot along with a question to get more context.

Limitations of the AI analysis

The AI analysis is limited to the sources that is supplied with, so it only knows about Phenix, and it only knows what is in the documentation, the videos and newsletters, and the papers we have supplied.

AI tools like this one can also just make mistakes and give incorrect answers. This does not seem to happen too often with this tool, but you need to always be on alert when using it. Use the tool as a helper, don't expect it to always be right.

If a detail is missing in the documentation, the AI analysis will not know it.

Privacy in the AI analysis

If you use the AI analysis tool, your log files are sent to the Phenix server, and from there on to Cohere and Gemini. That means the data could potentially be used by Cohere and Google in any way that they use other AI data that is sent to them.

Note that access to Cohere and Gemini is non-commercial only.

Applying NCS to a molecule to generate all NCS copies with `apply_ncs`

`apply_ncs.html`

Author(s)

- `apply_ncs`: Tom Terwilliger

Purpose

If you have a single copy of your molecule placed in the crystal and you have a set of NCS operators defined in an `ncs_info.ncs_spec` file (such as that written by `phenix.find_ncs`), you can apply the operators to your molecule to generate all the NCS copies. This can be useful if you want to work on one molecule in your asymmetric unit and propagate it to the other copies. You can generate your `ncs_info.ncs_spec` file using `phenix.find_ncs` and your original model containing all the NCS copies, then apply it to any one of the individual molecules to regenerate the others.

Usage

How `apply_ncs` works:

`apply_ncs` assumes that you are supplying it with one copy of your molecule, placed in your unit cell, and that you are also supplying a `ncs_info.ncs_spec` file that contains all the NCS operators for your structure along with the approximate molecular centers where each operator applies. `apply_ncs` assumes that the NCS operator with a center closest to the center of the molecule you supply (including space-group symmetry) is the operator corresponding to your input molecule.

`apply_ncs` then calculates the transformation for your molecule to molecule 1, and then uses all the NCS operators to transform molecule 1 to each NCS copy.

Output files from `apply_ncs`

`apply_ncs.pdb`: A PDB file with your input molecule placed at all the NCS-related positions in your structure

Examples

Standard run of `apply_ncs`:

Running `apply_ncs` is easy. From the command-line you can type:

```
phenix.apply_ncs mol1.pdb my_ncs.ncs_spec
```

If `apply_ncs` chooses the wrong NCS copy to match up to your input molecule, you can tell it which one your input molecule is with `match_copy`:

```
phenix.apply_ncs mol1.pdb my_ncs.ncs_spec match_copy=2
```

`apply_ncs` will then assume that you are inputting ncs copy 2.

Possible Problems

Specific limitations and problems:

Literature

Additional information

The documentation for `find_ncs` describes how to find and apply NCS information for helical arrays, point-group symmetry, and non-point-group symmetry.

List of all available keywords

`apply_ncsjob_title` = None Job title in PHENIX GUI, not used on command line
`input_files`
`pdb_in` = None Input PDB file
`ncs_in` = None File with NCS operators to be applied to `pdb_in` This can be a `.ncs_spec` file from `find_ncs`, `simple_ncs_from_pdb` or `find_ncs_from_density` It can also be NCS formatted for resolve, including rotations, translations, and centers NOTE: NCS operators only apply to a limited region within the unit cell. The center of this region for molecule 1 (the one you enter with `pdb_in`) is specified by the NCS center for molecule 1. If your molecule in `pdb_in` is not near this, then it will be brought as close as possible using space-group symmetry (as defined in the CRYST1 record of `pdb_in`) before application of NCS. If your molecule is closer to the center for operator k, then it will instead be brought to that center and then transformed to all other locations
`map_in` = None CCP4 or MRC-style map file (required if `trim_overlapping=True`). Map must cover the entire molecule including NCS
`copies`
`output_file`
`target_output_format` = *None `pdb` `mmCIF` Desired output format (if possible). Choices are None (try to use input format), `pdb`, `mmCIF`. If output model does not fit in `pdb` format, `mmCIF` will be used. Default is `pdb`
`pdb_out` = `apply_ncs.pdb` Output PDB file
`log` = `apply_ncs.log` Output log file
`params_out` = `apply_ncs_params.eff` Parameters file to rerun `apply_ncs`
`write_refine_ncs_out` = False Write out `refine_ncs_out` file for refinement
`refine_ncs_out` = `apply_ncs_refine.eff` NCS formatted for refinement. This will put all of the input chains as reference and all of the ncs-related chains in selection records, one for each NCS copy applied.
`directories`
`temp_dir` = "temp_dir" Optional temporary work directory
`output_dir` = "" Output directory where files are to be written
`gui_output_dir` = None
`ncs_trim_overlapping` = False Trim pieces that overlap (including those overlapping by NCS). Requires a full-size (not boxed) map file (`ccp4/mrc`). Also requires resolution of the map.
`resolution` = None Map resolution (for `trim_overlapping`)
`match_copy` = None Force input model to be associated with NCS copy `match_copy` This can be useful if your molecule center is close to an NCS center other than the one it is associated with
`no_match_copy` = None Do not allow input model to be associated with this NCS copy `no_match_copy`. This can be useful if your molecule center is close to an NCS center other than the one it is associated with
`apply_operators_directly` = False You can apply the NCS matrices directly to an input model, without considering one of the NCS copies to be the identity for this model.
`max_copies` = None You can limit the number of NCS copies to fewer than the actual number
`start_copy` = None You can start the NCS copies with any available copy
`only_start_copy` = None Just use `start_copy` and do not try to arrange chain names
`unique_only` = None Just use unique part of `pdb_in`
`chain_names` = None You can specify the chain names to generate. Note: there must be exactly the same number of `chain_names` as total chains generated by `apply_ncs`. This includes any chains that are just waters or ligands.
`used_chain_id_list` = None You can specify chain names not to generate
`use_space_group_symmetry` = True Normally `apply_ncs` uses space-group symmetry read from your input PDB file to find the appropriate operator. To disable this say `use_space_group_symmetry=False`. This is what you should do if you are using space group P1 as a dummy space group for example.
`control_verbosity` = False Verbose output
`raise_sorry` = False Raise sorry if problems
`debug` = False Debugging output
`dry_run` = False Just read in and check parameter names
`resolve_command_list` = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double

quotes (",) and each individual command in single quotes (') like this:
resolve_command_list="'no_build' 'b_overall 23' ";

- apply_ncsjob_title = None Job title in PHENIX GUI, not used on command lineinput_files_pdb_in = None Input PDB file ncs_in = None File with NCS operators to be applied to pdb_in This can be a .ncs_spec file from find_ncs, simple_ncs_from_pdb or find_ncs_from_density It can also be NCS formatted for resolve, including rotations, translations, and centers NOTE: NCS operators only apply to a limited region within the unit cell. The center of this region for molecule 1 (the one you enter with pdb_in) is specified by the NCS center for molecule 1. If your molecule in pdb_in is not near this, then it will be brought as close as possible using space-group symmetry (as defined in the CRYST1 record of pdb_in) before application of NCS. If your molecule is closer to the center for operator k, then it will instead be brought to that center and then transformed to all other locationsmap_in = None CCP4 or MRC-style map file (required if trim_overlapping=True). Map must cover the entire molecule including NCS copies.output_filetarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.pdb_out = apply_ncs.pdb Output PDB filelog = apply_ncs.log Output log fileparams_out = apply_ncs_params.eff Parameters file to rerun apply_ncswrite_refine_ncs_out = False Write out refine_ncs_out file for refinementrefine_ncs_out = apply_ncs_refine.eff NCS formatted for refinement. This will put all of the input chains as reference and all of the ncs-related chains in selection records, one for each NCS copy applied.directoriestemp_dir = "temp_dir" Optional temporary work directory output_dir = "" Output directory where files are to be writtenGUI_output_dir = None ncstrim_overlapping = False Trim pieces that overlap (including those overlapping by NCS). Requires a full-size (not boxed) map file (ccp4/mrc). Also requires resolution of the map.resolution = None Map resolution (for trim_overlapping)match_copy = None Force input model to be associated with NCS copy match_copy This can be useful if your molecule center is close to an NCS center other than the one it is associated withno_match_copy = None Do not allow input model to be associated with this NCS copy no_match_copy. This can be useful if your molecule center is close to an NCS center other than the one it is associated withapply_operators_directly = False You can apply the NCS matrices directly to an input model, without considering one of the NCS copies to be the identity for this model.max_copies = None You can limit the number of NCS copies to fewer than the actual numberstart_copy = None You can start the NCS copies with any available copyonly_start_copy = None Just use start_copy and do not try to arrange chain namesunique_only = None Just use unique part of pdb_inchain_names = None You can specify the chain names to generate. Note: there must be exactly the same number of chain_names as total chains generated by apply_ncs. This includes any chains that are just waters or ligands.used_chain_id_list = None You can specify chain names not to generateuse_space_group_symmetry = True Normally apply_ncs uses space-group symmetry read from your input PDB file to find the appropriate operator. To disable this say use_space_group_symmetry=False. This is what you should do if you are using space group P1 as a dummy space group for example.controlverbose = False Verbose outputraise_sorry = False Raise sorry if problemsdebug = False Debugging outputdry_run = False Just read in and check parameter namesresolve_command_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (",) and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' ";
- job_title = None Job title in PHENIX GUI, not used on command line
- input_files_pdb_in = None Input PDB file ncs_in = None File with NCS operators to be applied to pdb_in This can be a .ncs_spec file from find_ncs, simple_ncs_from_pdb or find_ncs_from_density It can also be NCS formatted for resolve, including rotations, translations, and centers NOTE: NCS operators only apply

to a limited region within the unit cell. The center of this region for molecule 1 (the one you enter with `pdb_in`) is specified by the NCS center for molecule 1. If your molecule in `pdb_in` is not near this, then it will be brought as close as possible using space-group symmetry (as defined in the CRYST1 record of `pdb_in`) before application of NCS. If your molecule is closer to the center for operator k, then it will instead be brought to that center and then transformed to all other locations `map_in = None` CCP4 or MRC-style map file (required if `trim_overlapping=True`). Map must cover the entire molecule including NCS copies.

- `pdb_in = None` Input PDB file
- `ncs_in = None` File with NCS operators to be applied to `pdb_in` This can be a `.ncs_spec` file from `find_ncs`, `simple_ncs_from_pdb` or `find_ncs_from_density` It can also be NCS formatted for resolve, including rotations, translations, and centers NOTE: NCS operators only apply to a limited region within the unit cell. The center of this region for molecule 1 (the one you enter with `pdb_in`) is specified by the NCS center for molecule 1. If your molecule in `pdb_in` is not near this, then it will be brought as close as possible using space-group symmetry (as defined in the CRYST1 record of `pdb_in`) before application of NCS. If your molecule is closer to the center for operator k, then it will instead be brought to that center and then transformed to all other locations
- `map_in = None` CCP4 or MRC-style map file (required if `trim_overlapping=True`). Map must cover the entire molecule including NCS copies.
- `output_filetarget_output_format = *None` `pdb` `mmCIF` Desired output format (if possible). Choices are `None` (try to use input format), `pdb`, `mmCIF`. If output model does not fit in `pdb` format, `mmCIF` will be used. Default is `pdb`. `pdb_out = apply_ncs.pdb` Output PDB file `log = apply_ncs.log` Output log file `params_out = apply_ncs_params.eff` Parameters file to rerun `apply_ncs` `write_refine_ncs_out = False` Write out `refine_ncs_out` file for refinement `refine_ncs_out = apply_ncs_refine.eff` NCS formatted for refinement. This will put all of the input chains as reference and all of the ncs-related chains in selection records, one for each NCS copy applied.
- `target_output_format = *None` `pdb` `mmCIF` Desired output format (if possible). Choices are `None` (try to use input format), `pdb`, `mmCIF`. If output model does not fit in `pdb` format, `mmCIF` will be used. Default is `pdb`.
- `pdb_out = apply_ncs.pdb` Output PDB file
- `log = apply_ncs.log` Output log file
- `params_out = apply_ncs_params.eff` Parameters file to rerun `apply_ncs`
- `write_refine_ncs_out = False` Write out `refine_ncs_out` file for refinement
- `refine_ncs_out = apply_ncs_refine.eff` NCS formatted for refinement. This will put all of the input chains as reference and all of the ncs-related chains in selection records, one for each NCS copy applied.
- `directorytemp_dir = "temp_dir"` Optional temporary work directory `output_dir = ""` Output directory where files are to be written `gui_output_dir = None`
- `temp_dir = "temp_dir"` Optional temporary work directory
- `output_dir = ""` Output directory where files are to be written
- `gui_output_dir = None`
- `ncstrim_overlapping = False` Trim pieces that overlap (including those overlapping by NCS). Requires a full-size (not boxed) map file (`ccp4/mrc`). Also requires resolution of the map. `resolution = None` Map resolution (for `trim_overlapping`) `match_copy = None` Force input model to be associated with NCS copy `match_copy` This can be useful if your molecule center is close to an NCS center other than the one it is associated with `no_match_copy = None` Do not allow input model to be associated with this NCS copy

no_match_copy. This can be useful if your molecule center is close to an NCS center other than the one it is associated with.

apply_operators_directly = False You can apply the NCS matrices directly to an input model, without considering one of the NCS copies to be the identity for this model.

max_copies = None You can limit the number of NCS copies to fewer than the actual number.

start_copy = None You can start the NCS copies with any available copy.

only_start_copy = None Just use start_copy and do not try to arrange chain names.

unique_only = None Just use unique part of pdb_in.

chain_names = None You can specify the chain names to generate. Note: there must be exactly the same number of chain_names as total chains generated by apply_ncs. This includes any chains that are just waters or ligands.

used_chain_id_list = None You can specify chain names not to generate.

use_space_group_symmetry = True Normally apply_ncs uses space-group symmetry read from your input PDB file to find the appropriate operator. To disable this say use_space_group_symmetry=False. This is what you should do if you are using space group P1 as a dummy space group for example.

- trim_overlapping = False Trim pieces that overlap (including those overlapping by NCS). Requires a full-size (not boxed) map file (ccp4/mrc). Also requires resolution of the map.
- resolution = None Map resolution (for trim_overlapping)
- match_copy = None Force input model to be associated with NCS copy match_copy. This can be useful if your molecule center is close to an NCS center other than the one it is associated with.
- no_match_copy = None Do not allow input model to be associated with this NCS copy no_match_copy. This can be useful if your molecule center is close to an NCS center other than the one it is associated with.
- apply_operators_directly = False You can apply the NCS matrices directly to an input model, without considering one of the NCS copies to be the identity for this model.
- max_copies = None You can limit the number of NCS copies to fewer than the actual number.
- start_copy = None You can start the NCS copies with any available copy.
- only_start_copy = None Just use start_copy and do not try to arrange chain names.
- unique_only = None Just use unique part of pdb_in.
- chain_names = None You can specify the chain names to generate. Note: there must be exactly the same number of chain_names as total chains generated by apply_ncs. This includes any chains that are just waters or ligands.
- used_chain_id_list = None You can specify chain names not to generate.
- use_space_group_symmetry = True Normally apply_ncs uses space-group symmetry read from your input PDB file to find the appropriate operator. To disable this say use_space_group_symmetry=False. This is what you should do if you are using space group P1 as a dummy space group for example.
- controlverbose = False Verbose output.
- raise_sorry = False Raise sorry if problems.
- debug = False Debugging output.
- dry_run = False Just read in and check parameter names.
- resolve_command_list = None You can supply any resolve command here. NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' ";
- verbose = False Verbose output.
- raise_sorry = False Raise sorry if problems.
- debug = False Debugging output.
- dry_run = False Just read in and check parameter names.

- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`

Automated molecular replacement with AutoMR

automr.html

Author(s)

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- AutoMR Wizard: Tom Terwilliger, Laurent Storoni, Randy Read, and Airlie McCoy
- phenix.xtriage: Peter Zwart

Purpose

Purpose of the AutoMR Wizard

The AutoMR Wizard provides a convenient interface to Phaser molecular replacement and feeds the results of molecular replacement directly into the AutoBuild Wizard for automated model rebuilding.

The AutoMR Wizard begins with datafiles with structure factor amplitudes and uncertainties, a search model or models, and identifies placements of the search models that are compatible with the data.

Usage

The AutoMR wizard has been deprecated and is no longer available in the Phenix GUI, but it may still be run from the command-line, and from parameters files. Both versions are identical except in the way that they take commands from the user. See Using the PHENIX Wizards for details of how to run a Wizard.

Summary of inputs and outputs for AutoMR

Input data file. This file can be in most any format, and must contain either amplitudes or intensities and sigmas. You can specify what resolution to use for molecular replacement and separately what resolution to use for model rebuilding. If you specify "0.0" for resolution (recommended) then defaults will be used for molecular replacement (i.e. use data to 2.5Å if available to solve structure, then carry out rigid body refinement of final solution with all data) and all the data will be used for model rebuilding.

Contents of the asymmetric unit. PHASER needs to know what the total mass in the asymmetric unit is (i.e. not just the mass of the search models). You can define this either by specifying one or more protein or nucleic acid sequence files, or by specifying protein or nucleic acid molecular masses, and telling the Wizard how many copies of each are present.

Space groups to search. You can request that all space groups with the same point group as the one you start out with be searched, and the best one be chosen. If you select this option then the best space group will be used for model rebuilding in AutoBuild.

Ensembles to search for. AutoMR builds up a model by finding a set of good positions and orientations of one "ensemble", and then using each of those placements as starting points for finding the next ensemble, until all the contents of the asymmetric unit are found and a consistent solution is obtained. You can specify any number of different ensembles to search for, and you can search for any number of copies of each ensemble. The order of searching for ensembles makes a difference, but Phaser chooses a sensible default search order based on the size and assumed accuracy of the different ensembles. In difficult cases you could try permuting the search order.

Each ensemble can be specified by a single PDB file or a set of PDB files. The contents of one set of PDB files for an ensemble must all be oriented in the same way, as they will be put together and used as a group always in the molecular replacement process.

You will need to specify how similar you think each input PDB file that is part of an ensemble is to the structure that is in your crystal. You can specify either sequence identity, or expected rmsd. Note that if you use a homology model, you should give the sequence identity of the template from which the model was constructed, not the 100% identity of the model!

Output of AutoMR

Output files from AutoMR

When you run AutoMR the output files will be in a subdirectory with your run number:

AutoMR_run_1_/ # subdirectory with results

- A summary file listing the results of the run and the other files produced:

AutoMR_summary.dat # overall summary

- A warnings file listing any warnings about the run

AutoMR_warnings.dat # any warnings

- A file that lists all parameters and knowledge accumulated by the Wizard during the run (some parts are binary and are not printed)

AutoMR_Facts.dat # all Facts about the run

- Molecular replacement model, structure factors, and map coefficients:

MR.1.pdb

MR.1.mtz

The AutoMR wizard writes out MR.1.pdb and MR.1.mtz as well as output log files. The MR.1.pdb file will contain all the components of your MR solution. If there are multiple PDB files in an ensemble, the model with the lowest estimated rmsd is chosen to represent the whole ensemble and is written to MR.1.pdb. If there are multiple copies of a model, the chains are lettered sequentially A B C... The MR.1.mtz file contains the data from your input file to the full resolution available, as well as sigmaA-weighted 2Fo-Fc map coefficients based on the rigid-body-refined model.

Model rebuilding. After PHASER molecular replacement the AutoMR Wizard loads the AutoBuild Wizard and sets the defaults based on the MR solution that has just been found. You can use the default values, or you may choose to use 2Fo-Fc maps instead of density-modified maps for rebuilding, or you may choose to start the model-rebuilding with the map coefficients from MR.1.mtz.

How to run the AutoMR Wizard

Running the AutoMR Wizard is easy. For example, from the command-line you can type:

```
phenix.automr native.sca search.pdb RMS=0.8 mass=23000 copies=1
```

The AutoMR Wizard will find the best location and orientation of the search model search.pdb in the unit cell based on the data in native.sca, assuming that the RMSD between the correct model and search.pdb is about 0.8 Å, that the molecular mass of the true model is 23000 and that there is 1 copy of this model in the asymmetric unit. Once the AutoMR Wizard has found a solution, it will automatically call the AutoBuild Wizard and rebuild the model.

Components, copies, search models, and ensembles

- Your structure is composed of one or more components such as a 20Kd subunit with sequence seq-of-20Kd-subunit.
- There may be one or more copies of each component in your structure.
- You can search for the location(s) of a component with a search model that consists of a single structure or an ensemble of structures.

What the AutoMR wizard needs to run

In a simple case where you have one search model and are looking for N copies of this model in your structure, you need:

- a datafile name (native.sca or data=native.sca)
- a datafile name (native.sca or data=native.sca)
- a search model (search_model.pdb or coords=search_model.pdb)
- a search model (search_model.pdb or coords=search_model.pdb)
- (3) how similar the search model is to your structure (RMS=0.8 or identity=75)
- (4) information about the contents of the asymmetric unit: (mass=23000 or seq_file=seq.dat) and (copies=1)

It may be advantageous to search using an ensemble of similar structures, rather than a single structure. If you have an ensemble of search models to search for, then specify it as

```
coords="model_1.pdb" coords="model_2.pdb" coords="model_3.pdb"
```

In this case you need to give the RMS or identity for each model: identity='45 40 35'. Each of the models in the ensemble must be in the same orientation as the others, so that the ensemble of models can be placed as a group in the unit cell. You may also use phenix.enssembler to generate a single multi-model PDB file containing the entire ensemble. In this case you should specify a single overall RMS or identity for the ensemble.

Running from a parameters file

You can run phenix.automr from a parameters file. This is often convenient because you can generate a default one with:

```
phenix.automr --show_defaults > my_automr.eff
```

and then you can just edit this file to match your needs and run it with:

```
phenix.automr my_automr.eff
```

If you are searching for more than one ensemble, or if there is more than one component in the a.u., then use a parameters file and specify them like this (put all of this in a file like "my_mr.eff" and run it with "phenix.automr my_mr.eff"):

```
automr {
  ensemble {
    ensembleID = "mol1"
    copies_to_find = 1
    coords = mol1.pdb
    identity = None
    RMS = "0.85"
  }
}
```

```

ensemble {
  ensembleID = "mol2"
  copies_to_find = 1
  coords = mol2.pdb
  identity = None
  RMS = "0.90"
}
component {
  seq_file = "seq1.dat"
  component_type = *protein nucleic_acid
  mass = None
  component_copies = 1
}
component {
  seq_file = "seq2.dat"
  component_type = *protein nucleic_acid
  mass = None
  component_copies = 1
}
}

```

Specifying which columns of data to use from input data files

If one or more of your data files has column names that the Wizard cannot identify automatically, you can specify them yourself. You will need to provide one column "name" for each expected column of data, with "None" for anything that is missing.

For example, if your data file data.mtz has columns F SIGF then you might specify

```

data=data.mtz
input_label_string="F SIGF"

```

You can find out all the possible label strings in a data file that you might use by typing:

```

phenix.autosol display_labels=data.mtz # display all labels for data.mtz

```

You can specify many more parameters as well. See the list of keywords, defaults and descriptions at the end of this page and also general information about running Wizards at Using the PHENIX Wizards for how to do this. Some of the most common parameters are:

```

data=w1.sca # data file
model=coords.pdb # starting model
seq_file=seq.dat # sequence file

```

Examples

Standard AutoMR run with coords.pdb native.sca

Run AutoMR using coords.pdb as search model, native.sca as data, assume RMS between coords.pdb and true model is about 0.85 Å, the sequence of true model is seq.dat and there is 1 copy in the asymmetric unit:

```

phenix.automr coords.pdb native.sca RMS=0.85 seq.dat copies=1 \
  n_cycle_rebuild_max=2 n_cycle_build_max=2

```

Specifying data columns

Run AutoMR as above, but specify the data columns explicitly:

```

phenix.automr coords.pdb RMS=0.85 seq.dat copies=1 \
  data=data.mtz input_label_string="F SIGF" \
  n_cycle_rebuild_max=2 n_cycle_build_max=2

```

Note that the data columns are specified by a string that includes both F and SIGF : "F SIGF". The string must match some set of data labels that can be extracted automatically from your data file. You can find the possible values of this string as described above with

```
phenix.automr display_labels=data.mtz
```

Specifying a refinement file for AutoBuild

Run AutoMR as above, but specify a refinement file that is different from the file used for the MR search:

```
phenix.automr coords.pdb RMS=0.85 seq.dat copies=1 \
data=data.mtz input_label_string="F SIGF" \
input_refinement_file=refinement.mtz \
input_refinement_labels="FP SIGFP FreeR_flag" \
n_cycle_rebuild_max=2 n_cycle_build_max=2
```

Note that the commands `input_refinement_file` and `input_refinement_labels` are in the scope "autobuild_variables" . These commands and others with this prefix are passed on to AutoBuild.

AutoMR searching for 2 components

Run AutoMR on a structure with 2 components. Define the components of the asymmetric unit with sequence files (beta.seq and blip.seq) and number of copies of each component (1). Define the search models with PDB files and estimated RMS from true structures. This is all done by creating a parameters file with all the control information in it. Put all of this in a file like "my_mr.eff" and run it with "phenix.automr my_mr.eff":

```
automr {
  data = "w1.sca"
  build = False
  ensemble {
    ensembleID = "mol1"
    copies_to_find = 1
    coords = mol1.pdb
    identity = None
    RMS = "0.85"
  }
  ensemble {
    ensembleID = "mol2"
    copies_to_find = 1
    coords = mol2.pdb
    identity = None
    RMS = "0.90"
  }
  component {
    seq_file = "seq1.dat"
    component_type = *protein nucleic_acid
    mass = None
    component_copies = 1
  }
  component {
    seq_file = "seq2.dat"
    component_type = *protein nucleic_acid
    mass = None
    component_copies = 1
  }
}
```

Specifying molecular masses of 2 components

Run AutoMR as in the previous example, except specify the components of the asymmetric unit with molecular masses (30000 and 20000), and define the search models with PDB files and percent sequence

identity with the true structures (50% and 60%). This is again all done by creating a parameters file with all the control information in it. Put all of this in a file like "my_mr.eff" and run it with "phenix.automr my_mr.eff":

```
automr {
  data = "w1.sca"
  seq_file = seq.dat
  ensemble {
    ensembleID = "mol1"
    copies_to_find = 1
    coords = mol1.pdb
    identity = 50
  }
  ensemble {
    ensembleID = "mol2"
    copies_to_find = 1
    coords = mol2.pdb
    identity = 60
  }
  component {
    component_type = *protein nucleic_acid
    mass = 30000
    component_copies = 1
  }
  component {
    component_type = *protein nucleic_acid
    mass = 40000
    component_copies = 1
  }
  autobuild_variables{
    n_cycle_rebuild_max = 1
  }
}
```

AutoMR searching for 2 components, but specifying the orientation of one of them

Run AutoMR on a structure with 2 components. Define the components of the asymmetric unit with sequence files (beta.seq and blip.seq) and number of copies of each component (1). Define the search models with PDB files and estimated RMS from true structures. Define the orientation and position of one component. Define the number of copies to find for each component (0 for beta, which is fixed, 1 for blip). This is again all done by creating a parameters file with all the control information in it. Put all of this in a file like "my_mr.eff" and run it with "phenix.automr my_mr.eff":

```
automr {
  data = "w1.sca"
  seq_file = seq.dat
  ensemble {
    ensembleID = "mol1"
    copies_to_find = 1
    coords = mol1.pdb
    identity = 50
  }
  ensemble {
    ensembleID = "mol2"
    copies_to_find = 0
    coords = mol2.pdb
    identity = 60
  }
  component {
    component_type = *protein nucleic_acid
    mass = 30000
    component_copies = 1
  }
  component {
    component_type = *protein nucleic_acid
```

```

mass = 40000
component_copies = 1
}
autobuild_variables{
n_cycle_rebuild_max = 1
}
fixed_ensembles {
fixed_ensembleID_list="mol2"
fixed_euler_list = 199.84 41.535 184.15
fixed_frac_list = -0.49736 -0.15895 -0.28067
}
}

```

Note: you have to define an ensemble for the fixed molecule (mol2 in this example) and that you search for 0 copies of this molecule.

Possible Problems

Specific limitations and problems

The AutoBuild Wizard can build PROTEIN, RNA, or DNA, but it can only build one at a time. If your MR model contains more than one type of chain, then you will need to run AutoBuild separately from AutoMR and when you run AutoBuild, specify one of them with `input_lig_file_list` and the type of chain to build with `chain_type`:

```

input_lig_file_list=ProteinPartofMRmodel.pdb
chain_type=DNA

```

- The keywords "cell" and "sg" have been replaced with "unit_cell" and "space_group" to make the keywords the same as in other phenix applications.
- The syntax for searches with more than one ensemble and more than one component have changed in PHENIX version 1.4.
- If you use an ensemble as a search model, the output structure will contain just the first member of the ensemble, so you may wish to put the member that is likely to be the most similar to the true structure as the first one in your ensemble.
- AutoMR no longer is able to pass arbitrary commands to AutoBuild (this was discontinued to allow for a fixed set of inputs for AutoBuild).
- The AutoMR Wizard can take most settings of most space groups, however it can only use the hexagonal setting of rhombohedral space groups (eg., #146 R3:H or #155 R32:H), and it cannot use space groups 114-119 (not found in macromolecular crystallography) even in the standard setting due to difficulties with the use of `asuset` in the version of `ccp4` libraries used in PHENIX for these settings and space groups.

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Carbohydrate Primer

carbo_primer.html

Introduction

A carbohydrate is an organic compound with the empirical formula $C_m(H_2O)_n$ where m could be different from n . Synonym of saccharide. Saccharides have several characteristics that differentiate them. These include placement of carbonyl group length of the carbon chain sequence of chiral centres handedness of the molecule linear or cyclic

- placement of carbonyl group
- length of the carbon chain
- sequence of chiral centres
- handedness of the molecule
- linear or cyclic

Cyclic saccharides are classified as alpha or beta isomers based on the position of the OH on the anomeric carbon relative to the large group on the ring.

Protein Chemistry

If your protein was produced in an insect or yeast cell line, you should only have NAG and MAN sugars, otherwise you should look at both types of PDB files to see which fits the density better (if you have any density past the main, five sugar core).

MAN

Mannose is a common saccharide in protein chemistry. There are two linear isomers and two cyclic forms of each. The PDB code for each is different because of the different restraints required for each. A leading form of error in protein models with carbohydrates is the use of the three-letter code, MAN, which is for the alpha-D-mannose when that vast majority of units in saccharides are beta-D-mannose or BMA.

NAG

NAG represents the beta form (the most common) and NDG is the alpha form. NAG is interesting because it's a saccharide with a substitute attached to the C2 carbon.

Polysaccharides

Saccharides polymerise into tree like structures that covalently bind to proteins. The links between are generally between the anomeric carbon and an oxygen on the "preceding" units moving from the protein outwards on the chain. Consider the disaccharide, lactose.

Note that the left saccharide is linked with its anomeric carbon via an oxygen to the C4 carbon. The left unit is β -D-galactose and the right is glucose. The link is known as β -(1-4) because the "attaching" unit is β and the it links C1 to C4.

Protein links

The most common link to proteins is the n-link which attaches to ASN in the ASN-X-SER/THR motif.

Using the Phenix chatbot assistant

chatbot.html

Authors

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Purpose

The Phenix chatbot is designed to guide Phenix users through the steps in structure determination, to provide help with running Phenix programs and interpreting their output, to help with debugging problems, and as a reference for the theory behind Phenix tools. The chatbot can be accessed in the Phenix GUI with the Chat button on the toolbar.

How the chatbot works

The Phenix chatbot is based on Google NotebookLM. The notebook was supplied with over 600 pages of Phenix documentation, all the Phenix newsletters, most of the Phenix video tutorials, and many of the Phenix publications.

The chatbot can be started with this chat link. You have to log into a Google account (any Google account should work). This creates a new clean copy of the notebook.

You can ask any questions about how to use Phenix and how to interpret results.

This type of AI does not save or learn from your questions. If you close the chat or hit Refresh, it will be in the same state as it started in. The AI does answer in the context of previous questions or requests, so if you ask it to give you instructions suitable for the GUI, it will try to do that throughout the session.

What kind of questions you can ask the chatbot

You can ask the chatbot almost anything, from big-picture questions to details.

For example:

"How do I solve my structure?"

"I have an mtz file and a sequence file. What do I do next?"

"What is the difference between an mtz file and an mrc file?"

"Can I use AlphaFold with X-ray data?"

"What inputs do I need for autosol?"

"How do I run refinement?"

"Please create an atom selection string that will select all the main-chain atoms of the nucleic acid residues in chain A"

"Please tell me how to run autobuild on the command-line"

"How do I know if my autobuild run worked?"

"How do I tell if I have a good model? Is an R-value of 0.25 ok?"

"What might prevent me from solving my structure by sad?"

Using the chatbot in parallel with other AI tools

A great way to get the most help with Phenix is to use the Phenix chatbot alongside other AI tools. For example if you have a log file from a Phenix run in the GUI and you have enabled AI analysis of log files, there will usually be a button next to the log file button labelled "AI analysis". If you analyze your results with this tool you can then follow up by asking more questions in the chatbot. You can even paste some of the output from the AI analysis into the chatbot and ask the chatbot to help you interpret the results.

You can also go to Perplexity and upload your log file with the paperclip icon. Then you can ask "Is this a good result?" and it will analyze your log file and summarize what it says. You can even ask it to find other log files from the web that are similar. Once you have uploaded your log file, you can ask the AI agent, "What should I do next?" to get ideas about next steps.

Limitations of the chatbot

The chatbot is limited to the sources that it is supplied with, so it only knows about Phenix, and it only knows what is in the documentation, the videos and newsletters, and the papers we have supplied.

AI chatbots like this one can also just make mistakes and give incorrect answers. This does not seem to happen too often with this chatbot, but you need to always be on alert when using it. Use the chatbot as a helper, don't expect it to always be right.

If a detail is missing in the documentation, the chatbot will not know it. For example, the current version of the chatbot does not know that for the `phenix.map_correlations` tool you must supply a resolution if you are comparing a model and a map because that was not stated in the documentation.

While the chatbot can give generally-correct overviews of how to solve a structure, sometimes it can get some aspects mixed up. For example, it might say that density modification improves resolution slightly. While that may be true for cryoEM, density modification does not change the resolution in X-ray crystallography, it only improves the phase information.

Similarly, when the atom selection example above is given to the chatbot, it may give an answer like: "backbone and chain A", instead of "backbone and (not protein) and chain A". Further, the chatbot may suggest testing atom selections with `phenix.pdb_atom_selection`, but can miss the detail that `phenix.pdb_atom_selection` does not recognize the "backbone" smart selection string.

Importing structures from the PDB

cif_as_mtz.html

Unlike most other reflection formats, the CIF files used by the PDB to store experimental data are not supported for direct input to PHENIX. `phenix.cif_as_mtz` is a simple utility for importing these files, and can automatically identify and fix several errors common in the PDB, such as incorrectly labeled intensities in place of amplitudes. Unfortunately, many structures fail to parse properly, so this tool is not guaranteed to work in all cases.

Using the PHENIX GUI

Open the main GUI and click "Import CIF structure factors" under the "Reflection tools" category. Only the CIF file is required, but a PDB file will often be necessary to obtain the correct symmetry information. Click "Use model to guess data type" if you want PHENIX to do a sanity check on the labeled amplitudes (which may not really be amplitudes!).

If you know the PDB ID but haven't downloaded the necessary files yet, you can simply enter the ID into the box in the upper right corner and click "Run", and the PDB and CIF files will be fetched automatically from the RCSB web server.

Command-line use

The output of `phenix.cif_as_mtz` without arguments is show below:

```
Usage: phenix.cif_as_mtz [reflection_cif_file] [options]

Options:
-h, --help show this help message and exit
--unit-cell=10,10,20,90,90,120|FILENAME
  External unit cell parameters
--space-group=P212121|FILENAME
  External space group symbol
--symmetry=FILENAME
  External file with symmetry information
--output-file-name=OUTPUT_FILE_NAME
  Output mtz file name.
--use-model=USE_MODEL
  Use PDB model to make better guess about reflection
  data type.
--wavelength-id=WAVELENGTH_ID
  Extract data set with given wavelength_id.
--crystal-id=CRYSTAL_ID
  Extract data set with given crystal_id.
--show-details-if-error
  Show data details for some errors.
--show-log
  Show some output.
--merge
  Merge non-unique data (preserves anomalous pairs)
--eliminate_sys_absent
  Remove systematic absences
--map_to_asu
  Map HKL indices to canonical asymmetric unit
--incompatible_flags_to_work_set
  When merging place reflections with incompatible
  flags into the working set.
```

```
Example: phenix.cif_as_mtz r109ksf.ent --symmetry=pdb109k.ent
```

The `--symmetry` argument is frequently required due to missing symmetry information; in these cases you should use the equivalent PDB file. Most of the other options are usually not necessary, but `--use-model` is

often helpful to get the data type right.

As a shortcut to download the files from the RCSB (or mirror) website, PHENIX includes a tool for fetching data for a given PDB code:

```
phenix.fetch_pdb 2hr0 phenix.fetch_pdb -x 2hr0
```

The argument "-x" specifies that the structure factors should be downloaded instead of the PDB file. (This will only work if the authors deposited their data.)

Output

Assuming that the data were successfully extracted, the program will create an MTZ file with columns FOBS, SIGFOBS (these will be treated as group by most applications in PHENIX), and R-free-flags. In many cases there will be fewer reflections in the output than data lines in the CIF file; this is due to the presence of non-numeric or otherwise unintelligible values.

List of references related to Phenix

[citations.html](#)

Current primary Phenix citation:

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The articles below include both papers produced by members of the Phenix collaboration, and works by external groups relating to methods used in Phenix (and elsewhere).

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Putting the best parts of several maps together with `combine_focused_maps`

`combine_focused_maps.html`

Author(s)

- `combine_focused_maps`: Tom Terwilliger

Purpose

The routine `combine_focused_maps` uses rigid-body refinement to place a model in several maps. It then uses the map-model correlation to identify which are the best parts of each map and the relationships among the models in the different maps to superpose the best parts of each map and create a composite map.

Usage

How `combine_focused_maps` works:

Combine-focused-maps uses the fit of chains in a model to each map to identify the correspondence between the maps. Using this approach it is not necessary that the maps are all superimposed or even that they have the same gridding or size.

If the maps all do superimpose (approximately), you can supply a single model. The model will be refined separately against each map with rigid-body refinement. The placements of each chain in each map will then be used to identify the transformations needed to superimpose the parts of each map corresponding to each chain.

If the maps are quite different, then you will need to supply one model for each map, where you have already placed each chain so that it superimposes with the appropriate part of each map. The models should all be the same except that they have been rigid-body-refined against the various maps. You can use the phenix tool `phenix.dock_in_map` to place the chains in each map if you have not done it in some other way.

Once a set of models has been set up to match the maps, the map-model correlation in the region of each chain in each model is calculated. This correlation is calculated in a special way: the B-values of all the atoms in the model are set to zero before map-model correlation is evaluated. This step is important in order to make sure that a perfect high-resolution map always will score better than a perfect low-resolution map. (If B-values are variable then a perfect high-resolution map and a perfect low-resolution map will both have correlations of 1, but with a B-value of zero the high-resolution map will have a higher correlation.)

For each chain in the model, the map-model correlations for each map are then used to identify the weighting on that map.

Alternatively, you can specify what parts of the model are to be grouped together for identifying which map to use in what region. You specify any number of `region_selection` selections (in Phenix atom selection syntax, like `(chain A and resseq 250:360)`). Each one is used to define a region in space and within that region the maps are weighted according to their correlations to the atoms selected with that selection.

An empirical weighting scheme is used. The relative weights of two maps (for this one chain or region) with a difference in `map_model_cc` of `delta_cc` are given by:

$$\exp(-\text{delta_cc}/\text{delta_cc_norm})$$

where `delta_cc_norm` is typically about 0.05. This means a map that has a map-model cc 0.1 less than another map gets a weight of about 1/10 of the better-fitting map. Using this weighting scheme and transformations calculated from the positions of this chain after rigid-body refinement against each map, a weighted average map is created for each chain in the model. These maps are all superimposed on the reference map and masked around the chain that each is to represent.

Finally all the weighted average maps are combined to form a composite map.

Normalization and scaling of input maps

With the default value of the keyword `normalize` (True), maps will be normalized by subtracting their mean value and scaling to yield an rms value of 1.0.

If values of `map_scale` are supplied (one per map; cannot duplicate values so use 0.9999, 1.000, and 1.0001 to if you want the same value of 1 for three maps), the maps will be scaled by these values after normalization.

If you want to just scale and not normalize, set `normalize` to False. (This was the behaviour before phenix version 5705)

Additional output files

If you specify `get_contribution_maps=True` (this is the default), at the end of the procedure you will get one map written out for each input map file, called something like "contribution_map_1.ccp4". This map has values from 0 to 1 showing the contribution of this input map file to the final map. You can use it to color your final composite map, showing how much of the information at each point came from this input map file.

If you specify `get_cc_map=True` (also default) you will get a map file called `cc_map.ccp4` that shows the map correlation between the model and the final composite map. The correlation map will be calculated over whatever regions in the model were used in forming the composite map (i.e. if you used individual chains, then the correlations will be calculated over each chain).

Examples

Standard run of `combine_focused_maps`:

Running `combine_focused_maps` is easy. From the command-line you can type:

```
phenix.combine_focused_maps reference_map.map focused_map_1.map \  
    focused_map_2.map model.pdb resolution=4
```

where `reference_map` is the map (CCP4, mrc or other related format) that will be used as the template to superimpose all other maps will be superimposed, and `focused_map_1.map` and `focused_map_2.map` are maps that are focused on some part of the map (and the remainder of those maps may be poor). The model will be rigid-body refined against all three maps (keeping chains fixed as rigid bodies). The best parts of each map will be selected and combined to create a composite map.

Possible Problems

Specific limitations and problems:

If the density in some of your maps is rather poor, the rigid-body refinement step may not work well. One thing to try in this case is to refine with `rigid_body_refinement_single_unit=True`. This will tend to hold everything together (but it won't give you an adjustment of individual chain positions). Another option is to refine (with rigid-body refinement only) the models before using `combine_focused_maps`. That way you have full control over the refinement process. Then in `combine_focused_maps` use `rigid_body_refinement=False`.

You cannot use `region_selection` with `use_model_symmetry`.

Literature

Additional information

If you supply a model for each map, the models are used to define what parts of each map are included in the analysis. If the model for your target map has chains ABC and one focused map has just chain A, then only chain A from the focused map will be transferred to the target map. If the model for the focused map has chains ABC then weighted versions of the map in the vicinities of ABC will be included.

If you have a model with symmetry (say, chains A B C D are all the same) and just one focused map (say, focused on A), then you can supply your target map, target model with chains ABCD, an one focused map and a model for the focused map that contains just chain A. Then you add the keyword `"use_model_symmetry=True"` and the focused map from chain A is applied to A B C and D in the target map.

If you supply half-maps corresponding to each full map, the half-maps will be processed in the same way as the full map, yielding one pair of half-maps at the end of the procedure. You can use these half-maps to estimate the local resolution of the final composite map (using `phenix.local_resolution`) or for other procedures requiring half maps. Note that the half maps could have discontinuities so they may not be appropriate for all uses of half maps.

List of all available keywords

`job_title = None` Job title in PHENIX GUI, not used on command line
`input_filesmap_file = None` Files with CCP4-style maps. May have origin in any location. All maps will be superimposed on the first map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement.
`map_scale = None` Scale to apply to map before use. Must be input in same order as `map_files`. If two values are the same, you need to make them slightly different (like 1 -> 1.000001). Applied after `normalize_maps`.
`model_file = None` Model placed in first map to be used to align maps. Each chain in the model will be used to mark an independent region in each map. Optionally one model can be supplied to match each map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement. NOTE: only the parts of maps marked by a chain in the corresponding model file will be included in averaging.
`keep_ligands = False` Allow ligands in density analysis. Note: it is possible that refinement will fail with some ligands. If so, please refine your model against each map individually first, supply those models, and turn off rigid-body refinement.
`half_map_1_file = None` Half-map 1 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files.
`half_map_2_file = None` Half-map 2 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files.
`region_selection = None` Model selection string defining one region to be considered. This is an alternative approach to defining regions (default is to use

each chain as a region). Any number of regions can be supplied. `output_files` `composite_map_file` = `composite_map.ccp4` Output map file with composite map `output_scale_factor` = `None` Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor. `refined_model_file` = `composite_map.pdb` Output model file after rigid-body refinement `composite_half_map_1_file` = `composite_half_map_1.ccp4` Output map file with composite half-map 1 `composite_half_map_2_file` = `composite_half_map_2.ccp4` Output map file with composite half-map 2 `crystal_info` `resolution` = `None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. `short_caption` = `High-resolution limit` `scattering_table` = `n_gaussian` `wk1995` `it1992` *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`. `superpose_model_symmetry` = `False` If `use_model_symmetry` = `True` then find all the chains in the target model (the first model supplied) that have the same sequence (or use `symmetry_chains` to ID these chains). Any chains in other models that match one of these chains will be matched to all of them. Using this keyword a single focused map can be applied to all corresponding chains in the target map/model. Note: cannot be combined with `region_selection`. `symmetry_chains` = `None` You can specify the chains that are to be considered equivalent. See `use_model_symmetry` for description. Note: cannot be combined with `region_selection`. `rigid_body_refinement` = `True` Run rigid-body refinement on input model vs each map `rigid_body_refinement_single_unit` = `False` Run rigid-bo...

- `job_title` = `None` Job title in PHENIX GUI, not used on command line

- `input_files` `map_file` = `None` Files with CCP4-style maps. May have origin in any location. All maps will be superimposed on the first map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement. `map_scale` = `None` Scale to apply to map before use. Must be input in same order as `map_files`. If two values are the same, you need to make them slightly different (like 1 -> 1.000001). Applied after `normalize_maps`. `model_file` = `None` Model placed in first map to be used to align maps. Each chain in the model will be used to mark an independent region in each map. Optionally one model can be supplied to match each map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement. NOTE: only the parts of maps marked by a chain in the corresponding model file will be included in averaging. `keep_ligands` = `False` Allow ligands in density analysis. Note: it is possible that refinement will fail with some ligands. If so, please refine your model against each map individually first, supply those models, and turn off rigid-body refinement. `half_map_1_file` = `None` Half-map 1 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files. `half_map_2_file` = `None` Half-map 2 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files. `region_selection` = `None` Model selection string defining one region to be considered. This is an alternative approach to defining regions (default is to use each chain as a region). Any number of regions can be supplied.

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- `output_files`
`composite_map_file = composite_map.ccp4` Output map file with composite
`mapoutput_scale_factor = None` Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.
`refined_model_file = composite_map.pdb` Output model file after rigid-body refinement
`composite_half_map_1_file = composite_half_map_1.ccp4` Output map file with composite half-map 1
`composite_half_map_2_file = composite_half_map_2.ccp4` Output map file with composite half-map 2
- `composite_map_file = composite_map.ccp4` Output map file with composite map
- `output_scale_factor = None` Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.
- `refined_model_file = composite_map.pdb` Output model file after rigid-body refinement
- `composite_half_map_1_file = composite_half_map_1.ccp4` Output map file with composite half-map 1
- `composite_half_map_2_file = composite_half_map_2.ccp4` Output map file with composite half-map 2
- `crystal_info`
`resolution = None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. `.short_caption = High-resolution limit`
`scattering_table = n_gaussian` Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`.
- `resolution = None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. `.short_caption = High-resolution limit`
- `scattering_table = n_gaussian` Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`.

- `superposeuse_model_symmetry = False` If `use_model_symmetry = True` then find all the chains in the target model (the first model supplied) that have the same sequence (or use `symmetry_chains` to ID these chains). Any chains in other models that match one of these chains will be matched to all of them. Using this keyword a single focused map can be applied to all corresponding chains in the target map/model. Note: cannot be combined with `region_selection.symmetry_chains = None` You can specify the chains that are to be considered equivalent. See `use_model_symmetry` for description. Note: cannot be combined with `region_selection.rigid_body_refinement = True` Run rigid-body refinement on input model vs each map `rigid_body_refinement_single_unit = False` Run rigid-body refinement with just one unit (do not break up into chains) `remove_water = True` Remove waters from input files `local_weighting = False` Use local correlation to weight individual regions of maps. Size of local region determined by `local_residues`. Alternative is one weight for each chain of each map. `local_residues = 10` Number of residues to use in local correlations. `delta_cc_norm = 0.05` When scaling, weight a map or part of a map that has map cc of `delta_cc` less than another map by $\exp(-\text{delta_cc}/\text{delta_cc_norm})$ less than the other map. `mask_atoms_atom_radius = None` Radius for masking atoms when transforming map. Default is $\max(3, \text{resolution})$. `mask_smoothing_radius = None` Radius for smoothing mask. Default is 3 times the resolution `normalize_maps = True` Normalize maps (mean zero, sd 1) before use. Applied before `map_scale` option. `get_contribution_maps = True` Create map-like files showing contribution of each map `get_cc_map = True` Create map-like file showing map-model correlation at end `use_first_map_as_template_only = False` Use first map as template only, do not take any density from it

- `use_model_symmetry = False` If `use_model_symmetry = True` then find all the chains in the target model (the first model supplied) that have the same sequence (or use `symmetry_chains` to ID these chains). Any chains in other models that match one of these chains will be matched to all of them. Using this keyword a single focused map can be applied to all corresponding chains in the target map/model. Note: cannot be combined with `region_selection`.

- `symmetry_chains = None` You can specify the chains that are to be considered equivalent. See `use_model_symmetry` for description. Note: cannot be combined with `region_selection`.

- `rigid_body_refinement = True` Run rigid-body refinement on input model vs each map

- `rigid_body_refinement_single_unit = False` Run rigid-body refinement with just one unit (do not break up into chains)

- `remove_water = True` Remove waters from input files

- `local_weighting = False` Use local correlation to weight individual regions of maps. Size of local region determined by `local_residues`. Alternative is one weight for each chain of each map.

- `local_residues = 10` Number of residues to use in local correlations.

- `delta_cc_norm = 0.05` When scaling, weight a map or part of a map that has map cc of `delta_cc` less than another map by $\exp(-\text{delta_cc}/\text{delta_cc_norm})$ less than the other map.

- `mask_atoms_atom_radius = None` Radius for masking atoms when transforming map. Default is $\max(3, \text{resolution})$.

- `mask_smoothing_radius = None` Radius for smoothing mask. Default is 3 times the resolution

- `normalize_maps = True` Normalize maps (mean zero, sd 1) before use. Applied before `map_scale` option.

- `get_contribution_maps = True` Create map-like files showing contribution of each map

- `get_cc_map = True` Create map-like file showing map-model correlation at end

- `use_first_map_as_template_only = False` Use first map as template only, do not take any density from it

- controlnproc = 1 Number of processors to use random_seed = 171731 Random seed verbose = False Verbose output
- nproc = 1 Number of processors to use
- random_seed = 171731 Random seed
- verbose = False Verbose output
- guiGUI-specific parameter required for output directory output_dir = None restraint_files = None
- output_dir = None
- restraint_files = None